

EXHIBIT 30

the 'information' and 'communication' fields. The 'information' field is defined as:

...the study of the processes of information creation, organisation, storage, retrieval, dissemination and use, and the social, cultural, economic and political contexts in which these processes take place. (p. 1)

The 'communication' field is defined as:

...the study of the processes of communication, the social, cultural, economic and political contexts in which these processes take place, and the impact of communication on society. (p. 1)

The 'information science' field is defined as:

...the study of the processes of information creation, organisation, storage, retrieval, dissemination and use, and the social, cultural, economic and political contexts in which these processes take place, and the impact of information science on society. (p. 1)

The 'information studies' field is defined as:

...the study of the processes of information creation, organisation, storage, retrieval, dissemination and use, and the social, cultural, economic and political contexts in which these processes take place, and the impact of information studies on society. (p. 1)

The 'information technology' field is defined as:

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The 'information systems' field is defined as:

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The 'information management' field is defined as:

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The 'information policy' field is defined as:

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The 'information law' field is defined as:

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The 'information ethics' field is defined as:

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the 1990s, the number of people in the world who are under 15 years of age is expected to increase from 1.1 billion to 1.5 billion.

As the world's population grows, the demand for food and other resources will increase. This will put pressure on the environment and on the world's food supply.

One way to meet this demand is to increase the amount of food that is produced. This can be done by using more land for agriculture.

Another way to meet this demand is to increase the efficiency of food production. This can be done by using better farming techniques.

Both of these methods have their own problems. Increasing the amount of land used for agriculture can lead to deforestation and the loss of biodiversity.

Increasing the efficiency of food production can lead to the use of more pesticides and fertilizers, which can be harmful to the environment.

One solution is to use a combination of these methods. This would involve using better farming techniques on existing farmland.

Another solution is to use more land for agriculture, but in a way that is sustainable. This would involve using techniques that protect the environment.

There are many other ways to meet the world's growing demand for food. The key is to find a way that is sustainable and that protects the environment.

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the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1999. The public sector has become a major employer in the UK, and its growth has been a key factor in the overall growth of the economy.

The public sector has also become a major provider of social services, and its growth has been a key factor in the overall growth of the economy. The public sector has become a major provider of social services, and its growth has been a key factor in the overall growth of the economy.

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the 1990s, the number of people in the UK who are aged 65 and over has increased by 1.5 million, and the number of people aged 75 and over has increased by 1.1 million (Office for National Statistics 2000). The number of people aged 65 and over is projected to increase to 10.5 million by 2026, and the number of people aged 75 and over to 6.5 million (Office for National Statistics 2000).

There is a growing awareness of the need to address the needs of older people in the UK. The Department of Health (2000) has published a strategy for older people, which sets out the government's commitment to improve the lives of older people. The strategy is based on the following principles: (1) older people should be able to live independently and actively; (2) older people should be able to access the services and support they need; (3) older people should be able to participate in the decisions that affect their lives; and (4) older people should be able to live in a safe and secure environment.

The strategy is based on the following principles: (1) older people should be able to live independently and actively; (2) older people should be able to access the services and support they need; (3) older people should be able to participate in the decisions that affect their lives; and (4) older people should be able to live in a safe and secure environment. The strategy is based on the following principles: (1) older people should be able to live independently and actively; (2) older people should be able to access the services and support they need; (3) older people should be able to participate in the decisions that affect their lives; and (4) older people should be able to live in a safe and secure environment.

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AI quad-phase power modules designed for VPD

Learn more about Infineon's voltage regulation solutions for AI & data centers



OptiMOS™ quad-phase power modules 280 A in 10x9x5 mm package

Reducing the TCO of Generative AI data centers

Data centers are experiencing unprecedented growth primarily driven by increased investments in artificial intelligence (AI). Data centers are currently responsible for more than two percent of global energy consumption. Fueled by AI, the power demands within data centers are expected to grow by 165% between 2023 and 2030. Continually improving the efficiency and power densities of power conversion from grid-to-core is

vital to enable further advancements in compute performance while reducing total cost of ownership (TCO).

High density power module roadmap



Power modules portfolio overview

280 A

160 A

140 A

OptiMOS™ TDM2454xx quad-phase power modules 280 A target high-performance AI data centers. These modules enable true vertical power delivery (VPD) and offer industry's best power density of 2.0A/mm². TDM2454xx modules combine Infineon's robust OptiMOS™ 6 trench technology, a chip-embedded package with enhanced electrical and thermal efficiencies, and a new inductor technology to enable lower profile and therefore, true vertical power delivery. Additionally, the TDM2454xx has a footprint conducive to VPD systems enabling module tiling and improving current flow.

Key features:

- Industry-leading power density: 10x9x5 mm for 280 A peak
- Industry first quad-phase power module option with integrated embedded

capacitors available in 10x9x5mm package

- >2.0 A/mm² TDC (thermally managed)*
- First module optimized for VPD with a footprint designed for optimal for tiling arrays of modules
- Proprietary magnetics enable low-profile vertical power delivery in compact footprint

**Coldplate temperatures 80°C and below*

Download product brief



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TDM2454xx

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Mar 11, 2025



Life Cycle Assessment – 2nd stage soluti

Introduction to Vertical Power Delivery

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	Website feedback		



Powering AI Accelerators by Power Modules with Minimized Power Distribution Network Losses

Wenkang Huang, Ravidasa Hegde, Paul Yeaman,
Surendhar Somasundaram, Danny Clavette, Darryl Galipeau
Infineon Technologies Americas, U.S.A.



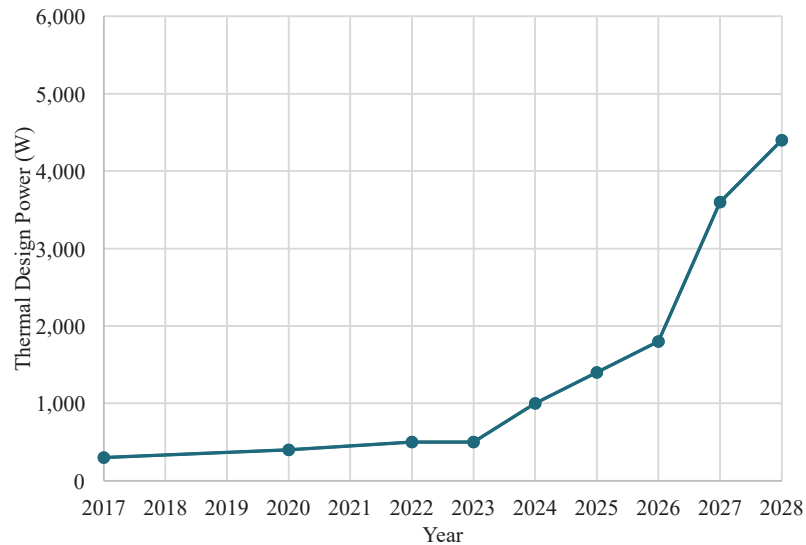
Presentation Number: T09.2

Outline

- Introduction – GPU Power Delivery Challenges
- Lateral Power Delivery to GPUs
- Vertical Power Delivery to GPUs
- Two Dual-Phase Synchronous Buck Converter Power Modules
- Evaluation of Lateral and Vertical Power Delivery Solutions
- Conclusion

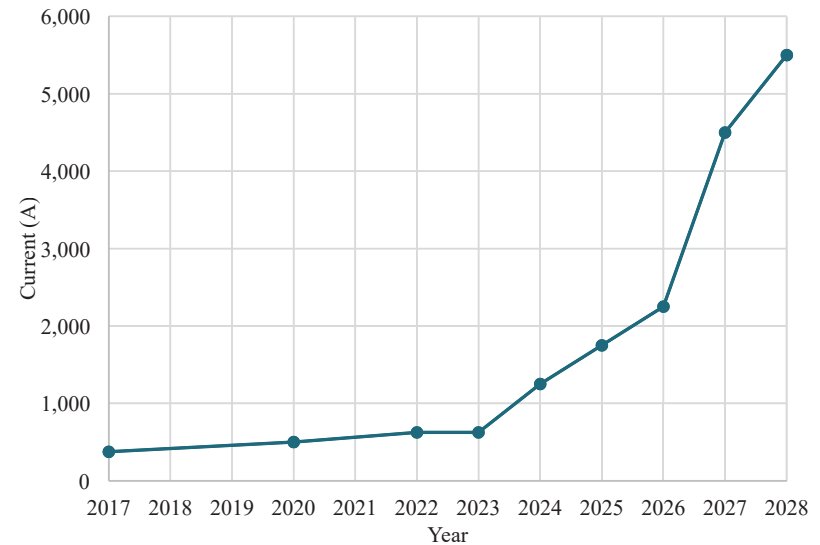
Introduction – NVIDIA GPU Power Consumption

GPU Power¹



Note 1: Correction to Figure 1 in the paper

GPU Current²



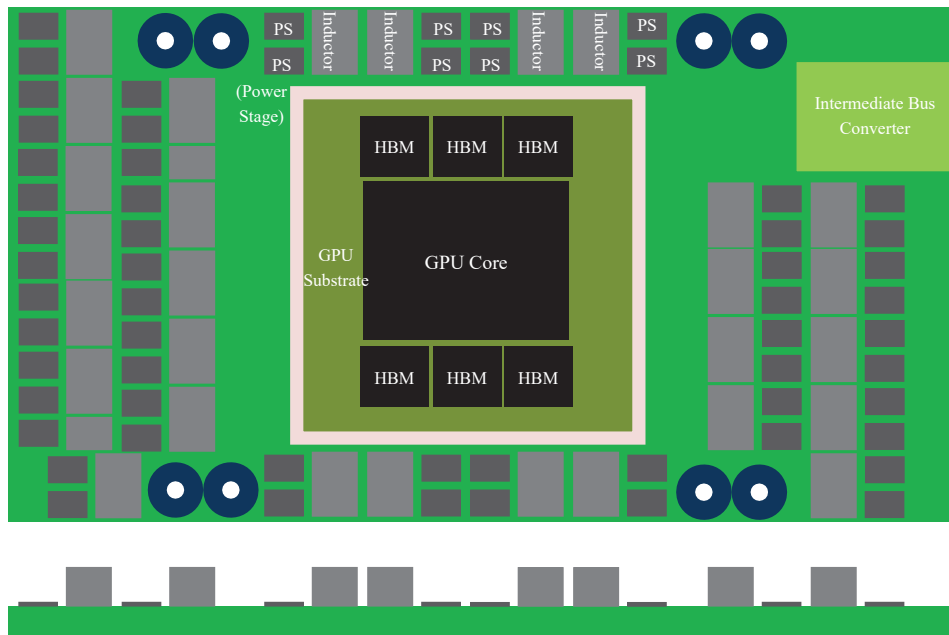
Note 2: Assuming Typical 0.8-V Core Voltage from 2017 to 2028



2nd Stage: Multiphase Synchronous Buck Converters

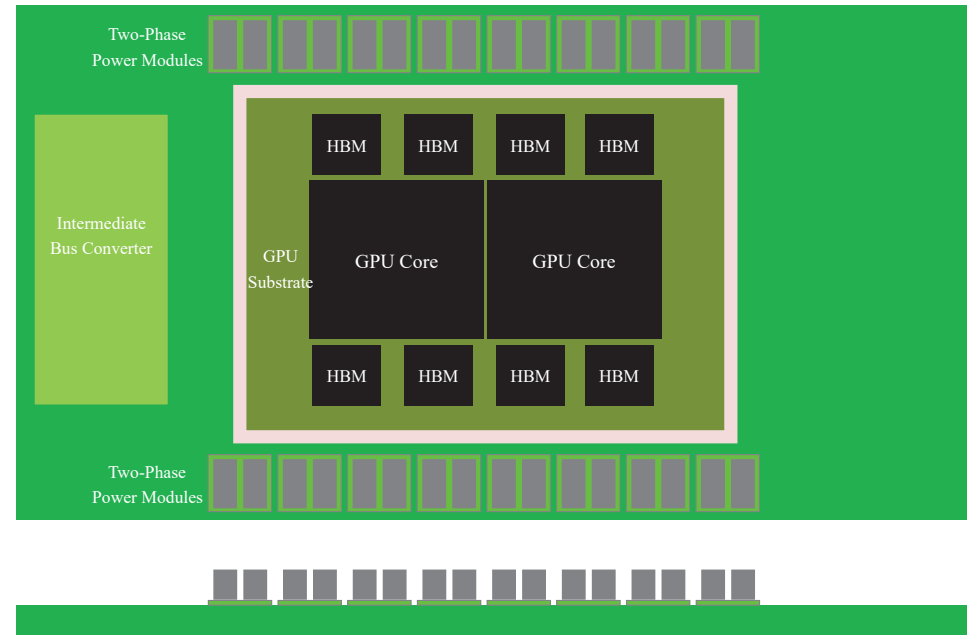
Lateral Power Delivery to GPUs and HBMs

Lateral Discrete Power Solution



2nd Stage: Power Stages + Dual-Phase Inductors

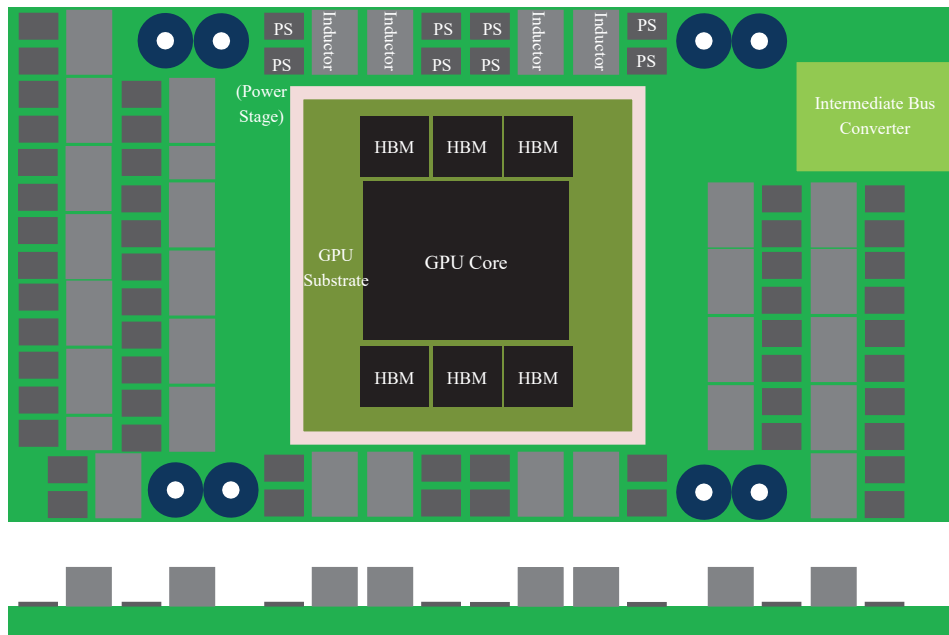
Lateral Module Power Solution



2nd Stage: Dual-Phase Modules on Topside

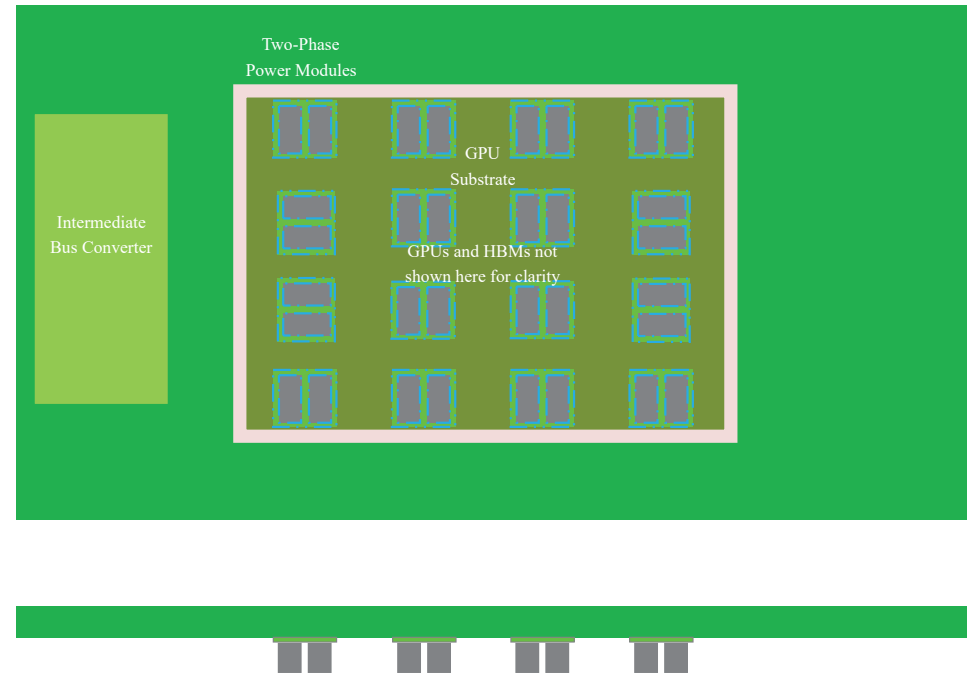
Lateral and Vertical Power Delivery to GPUs and HBMs

Lateral Discrete Power Solution



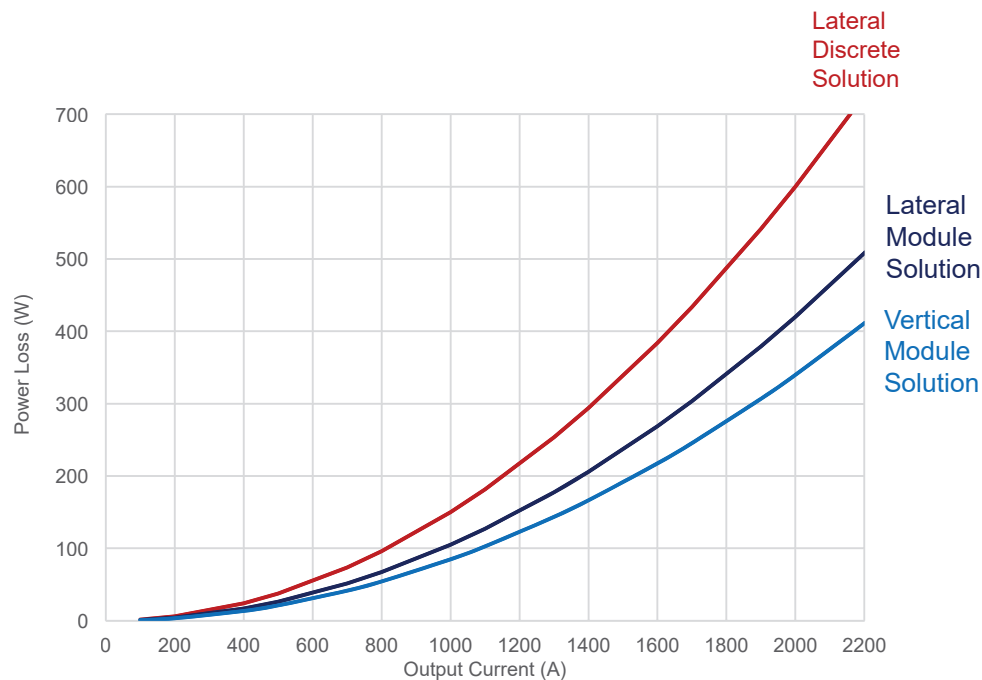
2nd Stage: Power Stages + Dual-Phase Inductors

Vertical Module Power Solution



2nd Stage: Dual-Phase Modules on Backside

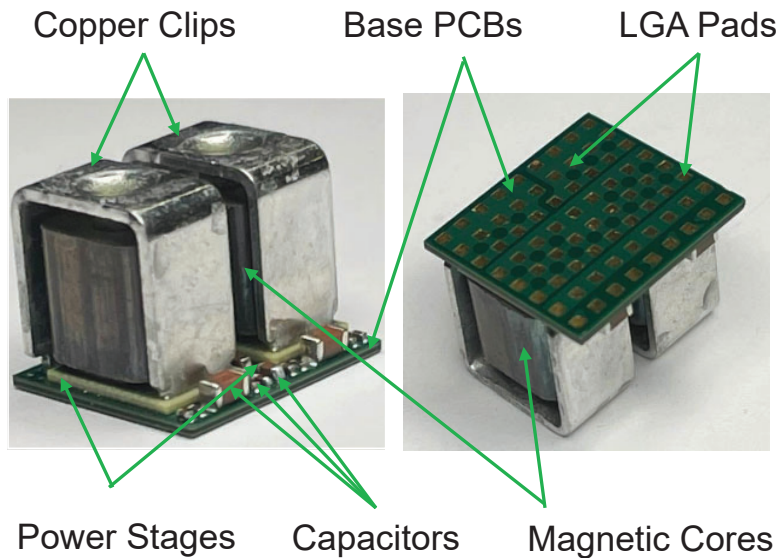
Prediction of PDN Loss with Different Power Solutions



- As GPU currents rise, PDN losses limit compute currents
- Limiting PDN loss is critical to maximize performance per Watt
 - Lower core voltage
 - Better transient response
 - Enable new compute architectures
- Vertical Power Delivery is necessary to minimize Total Cost of Ownership (TCO)
 - Less waste heat in PDNs and voltage regulators
 - More compute (GPUs) in AI factory footprint

Two-Phase Synchronous Buck Converter Modules

Module in 9 mm x 10 mm x 8 mm

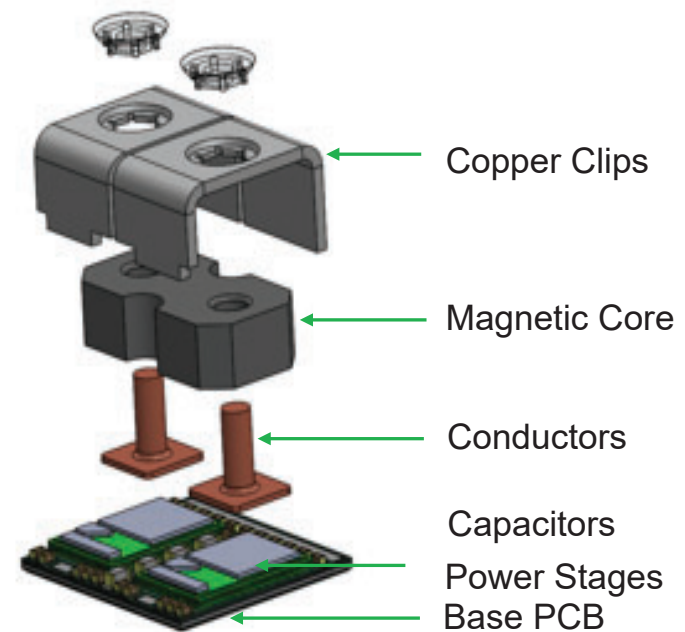


- TDC Current: **80 A**
- Peak Current: **140 A**
- Current Density: **> 1.5 A/mm²**

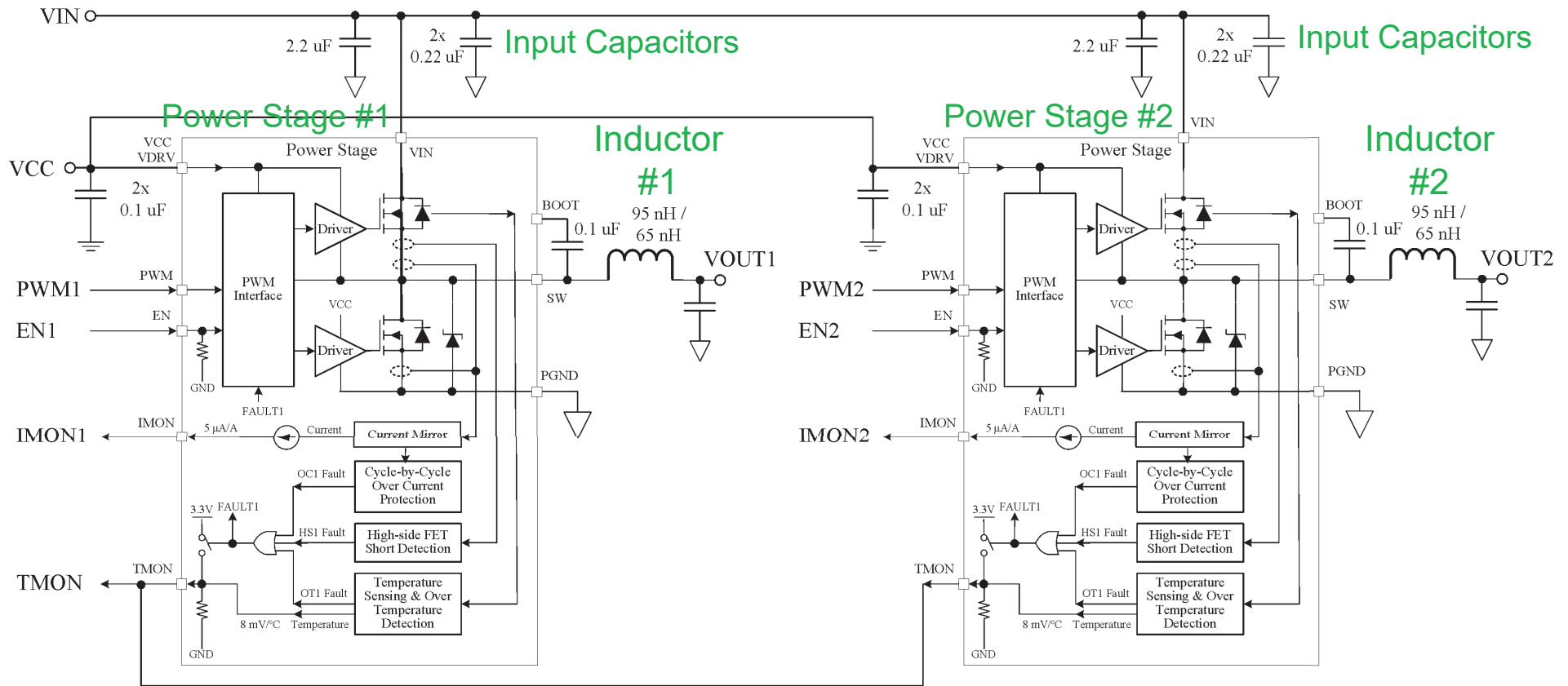
Benefits:

- Footprint Reduction
- Switching Noise Reduction

Module in 9 mm x 10 mm x 5 mm

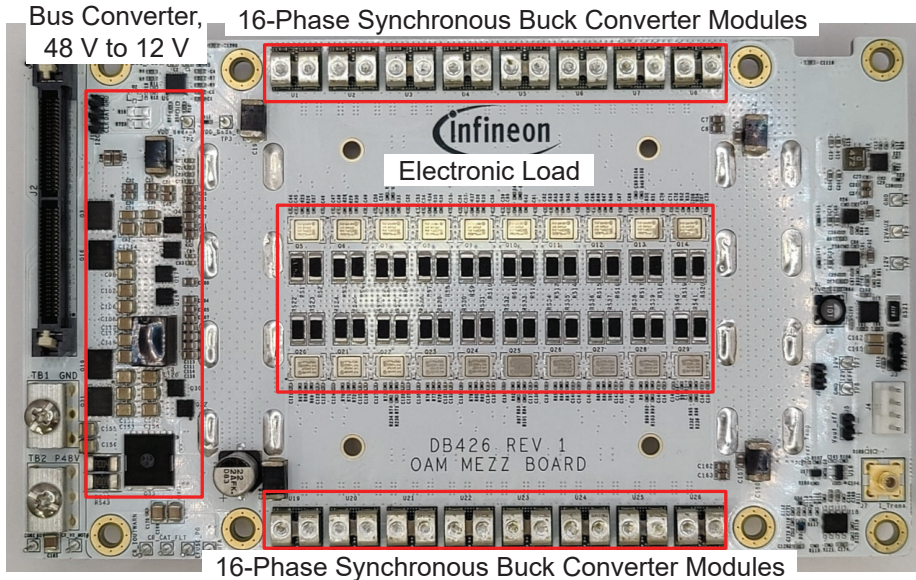


Block Diagram of Two-Phase Converter Modules

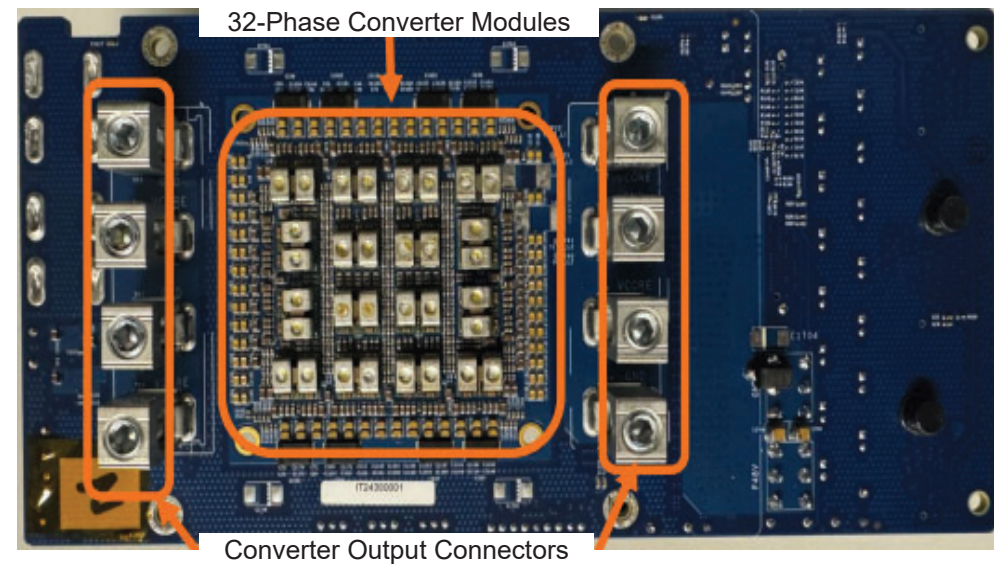


Test Boards of Lateral and Vertical Power Delivery

First Stage:
Intermediate
Bus Converter,
48 V to 12 V

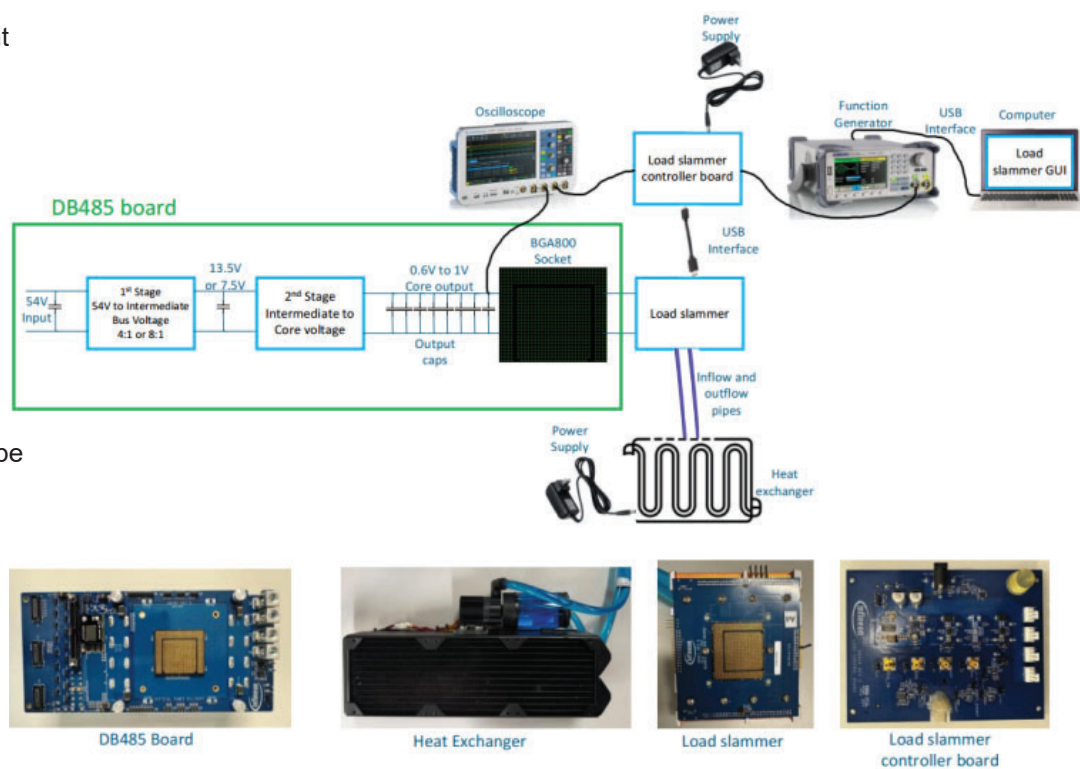
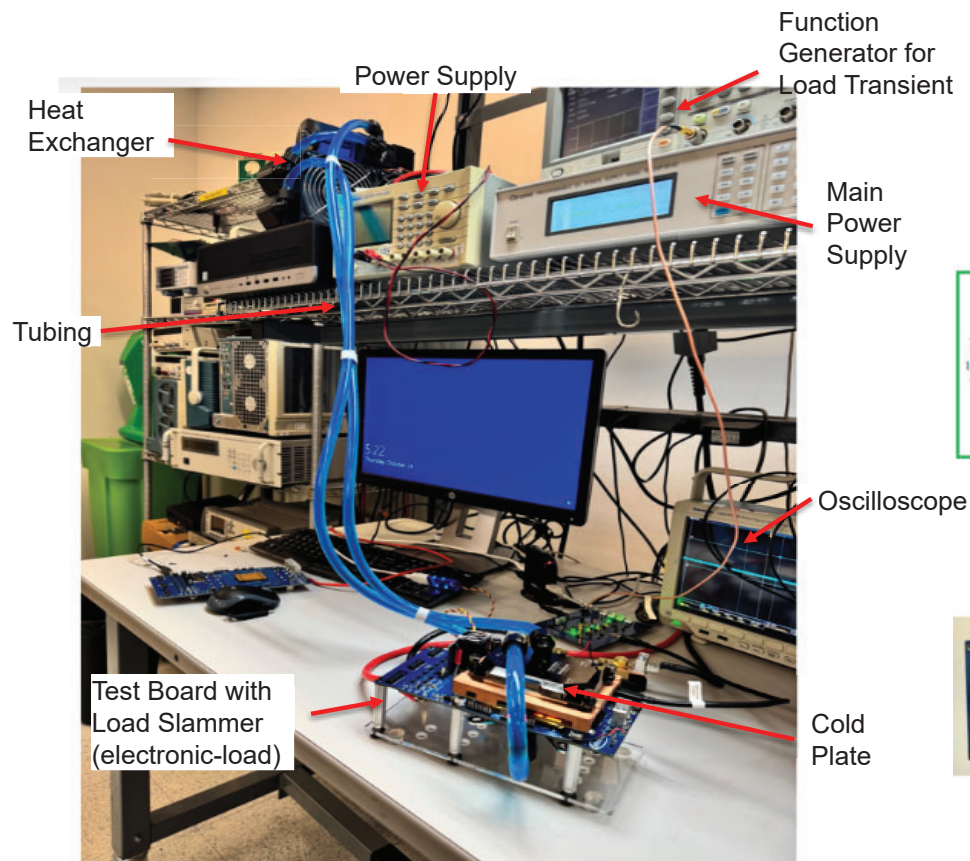


Lateral Power Delivery Board
with 5-mm Modules

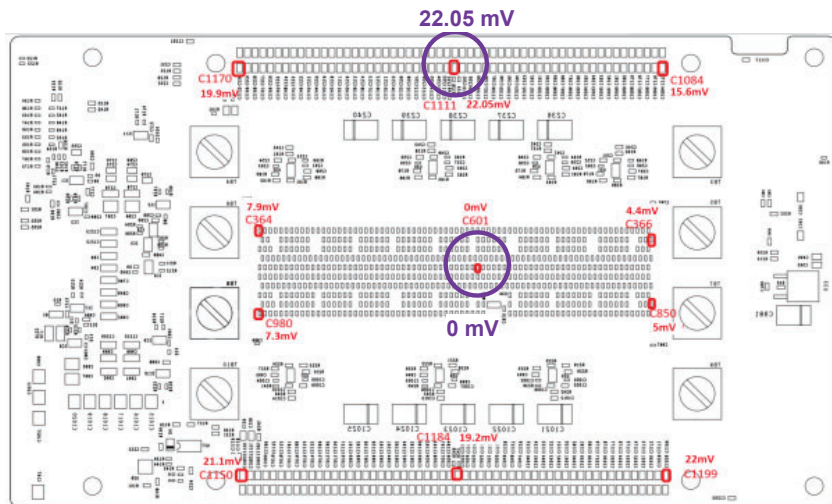


Vertical Power Delivery Board
with 5-mm Modules

Test Setup of Lateral & Vertical Power Delivery Solutions

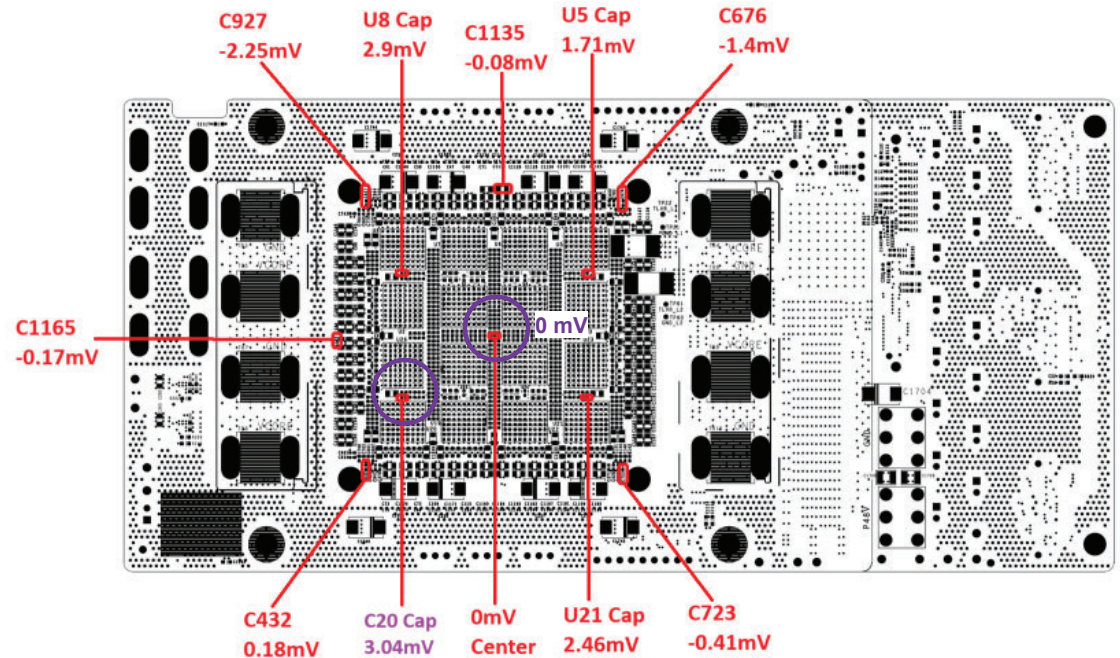


Comparison of Load Voltage Gradients



Lateral Power Delivery Board

Backside Load Voltage Distribution

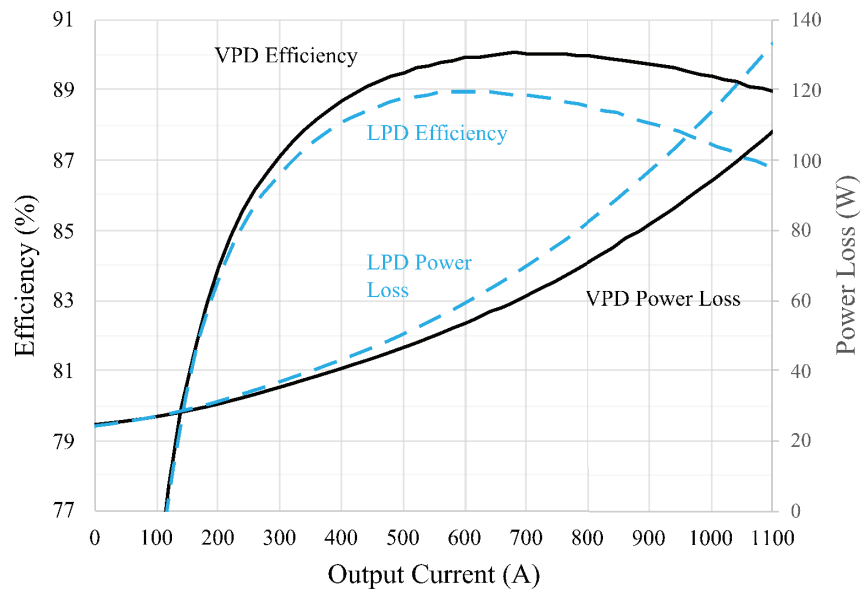


Vertical Power Delivery Board

Backside Load Voltage Distribution

Summary of Test Results

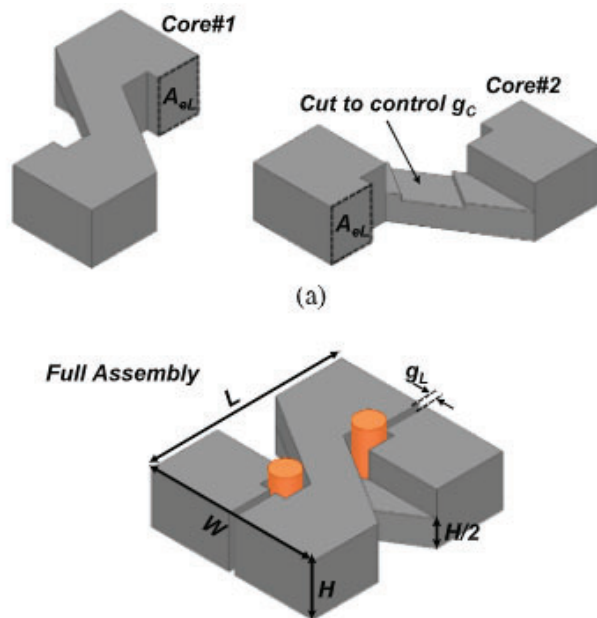
– Lateral Power Delivery (LPD) vs. Vertical Power Delivery (VPD)



	LPD Board	VPD Board	Improvement
Voltage Gradient in Load Area	22.05 mV	3.04 mV	19.01 mV ↓
Peak Efficiency / Full-Load (TDC) Efficiency at 1,100 A	88.9% / 86.7%	90.1% / 88.9%	Peak: 1.2% ↑ TDC: 2.2% ↑
Power Loss at 1,100-A TDC Current	132.9 W	108.4 W	TDC: 24.5 W ↓
Output Capacitance Required to Meet Same Load Transients	23.48 mF	16.06 mF	7.42 mF (31.6%) ↓

Converter Modules to Improve Transient Response

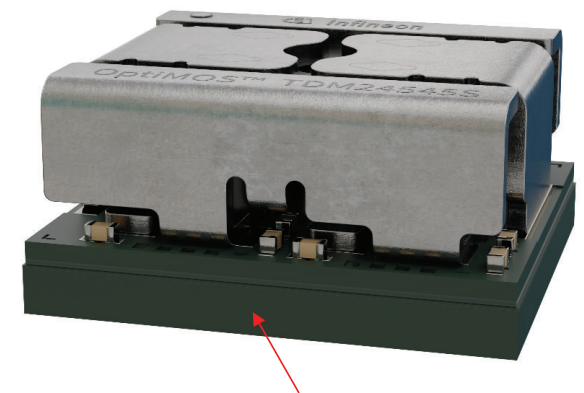
Two-Phase Twisted Core
Coupled Inductor¹ for Modules



Two-Phase Module² with
TLVR Inductors
(10 x 9 x 4.8 mm)



Four-Phase Module² with
Embedded Input / Output
Capacitors (10 x 9 x 5 mm)



TDM24545S Capacitor Layer

Note 1: A. Naradhipa, et. al., "Novel twisted core hybrid coupled inductor for vertical stack voltage regulator module," IEEE TPEL, Vol 41, No. 4, 2026

Note 2: <https://www.infineon.com/products/power/dc-dc-converters/high-density-power-modules>

Conclusion

- The next generation GPUs requires more than 5000-A current in three years and demands power solutions in limited areas.
- Power modules saves about 40% footprint area and reduces switching noise on the motherboard.
- Two-phase modules are developed for both lateral power delivery and vertical power delivery.
- Test of efficiency, power loss, voltage gradient, and transient response demonstrates benefits of vertical power delivery over lateral power delivery using multiple two-phase power modules.



APEC 2026

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PUBLIC VERSION

Powering the Future of AI: High Power Density Solutions for AI Accelerators

Surendhar Somasundaram

Paul Yeaman

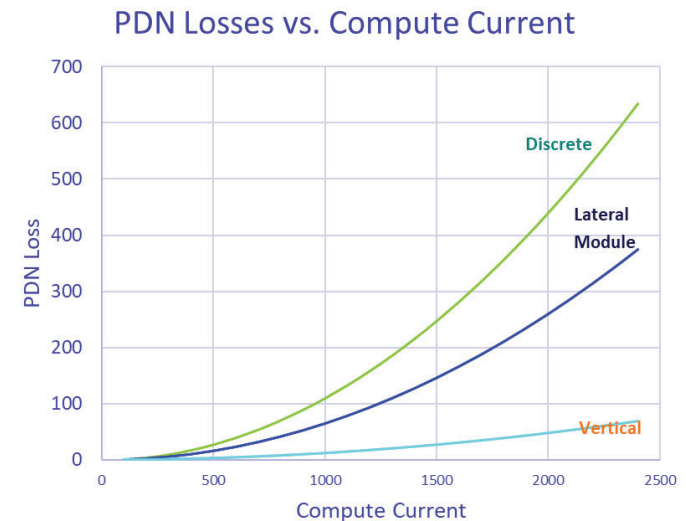


October 13-16, 2025
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Power Delivery Innovation : Motivation

- Compute systems are thermally limited
- As GPU currents rise, heat from PDN losses limit compute power
- Limiting PDN loss is critical to maximize compute power
 - Lower core voltage
 - Better transient response
 - Enable new compute architectures
 - Waferscale engines
- Vertical Power Delivery (VPD) is necessary to minimize PDN losses
 - Growing processor power requirements drive geometric PDN loss increase
 - VPD is required to keep pace with growing processor power demand



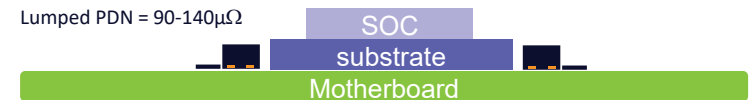
PUBLIC VERSION

Power Delivery Solutions for AI Accelerators

Discrete

- Lowest cost, with established eco-system and quality record
- PDN losses exceed 100W for GPU currents beyond 850-1000A

Lumped PDN = 90-140 $\mu\Omega$



Module (Lateral)

- Integrates power stage, inductor and capacitors into a single device, enabling more current to be sourced close to processor
- PDN losses exceed 100W for GPU currents beyond 1100 – 1350A

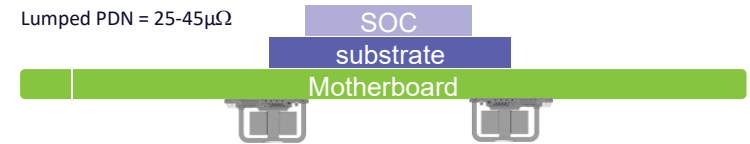
Lumped PDN = 55-80 $\mu\Omega$



Module (Backside Lateral)

- Relocates the module to the backside of the motherboard, shortening the high current path to GPU
- PDN losses exceed 100W for GPU currents beyond 1500 – 2000A

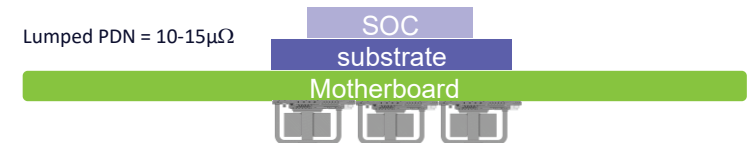
Lumped PDN = 25-45 $\mu\Omega$



Module (Vertical)

- Positions modules directly under processor, further shortening current path
- PDN losses exceed 100W for GPU currents beyond 2500 – 3000A

Lumped PDN = 10-15 $\mu\Omega$

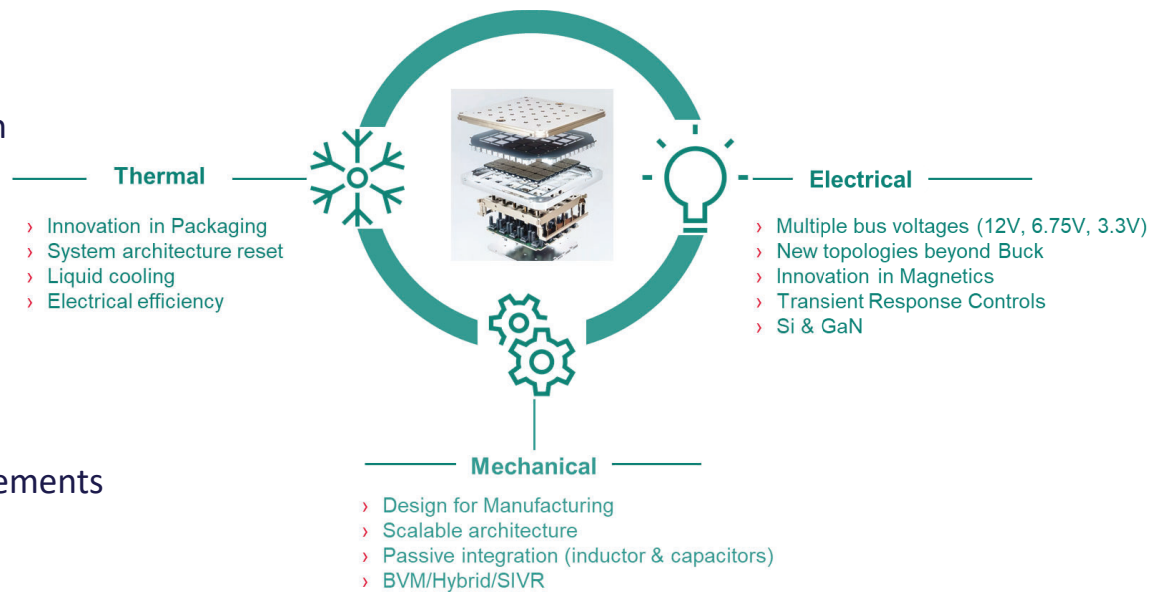


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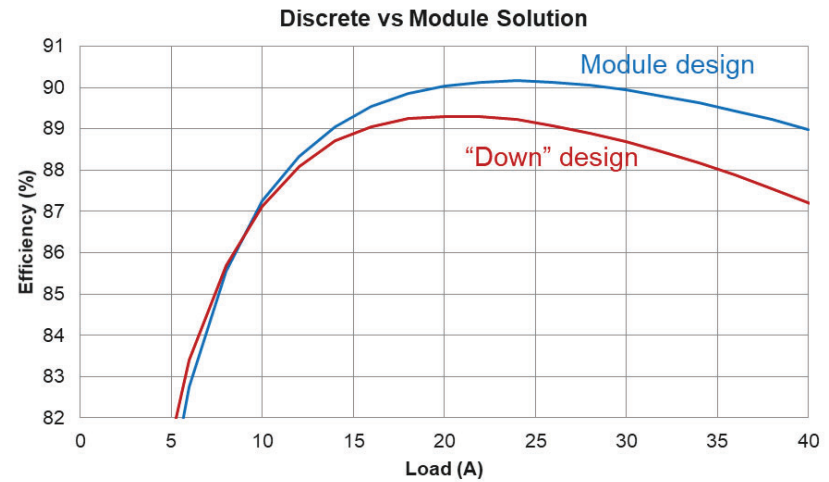
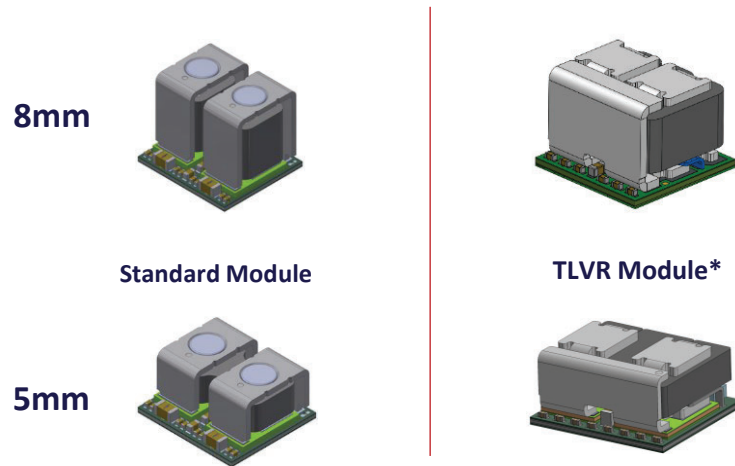
Impact of Vertical Power Delivery

- VR relocation under processor
 - Improves PDN losses
 - Allows for more powerful GPUs
 - Requires significant system redesign
- BVMs will continue to improve
 - Move closer to SOC
 - Improve power density
- Challenges will drive technology improvements
 - Electrical
 - Mechanical
 - Thermal

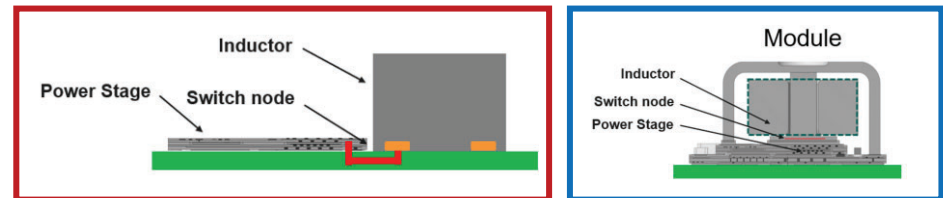


PUBLIC VERSION

Gen 1 Module (Lateral) : Dual Phase 160A, 10x9mm



- › Higher Efficiency than equivalent discrete solution
- › 160A peak, 80A continuous
- › Low Thermal Impedance $\theta_{\text{Junction-to-Top}} \sim 2.5\text{-}3.0^{\circ}\text{C/W}$
- › TLVR versions available in same form factor

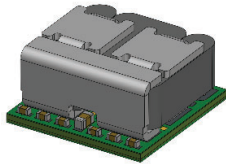


*TLVR denotes Trans Inductor Voltage Regulator

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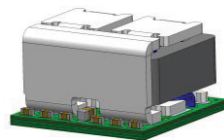
Gen 2 Module (Backside Lateral) : Dual Phase 160A, 8x8mm

4mm



Standard Module

5mm

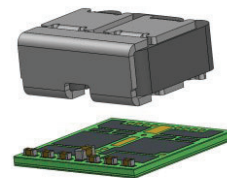
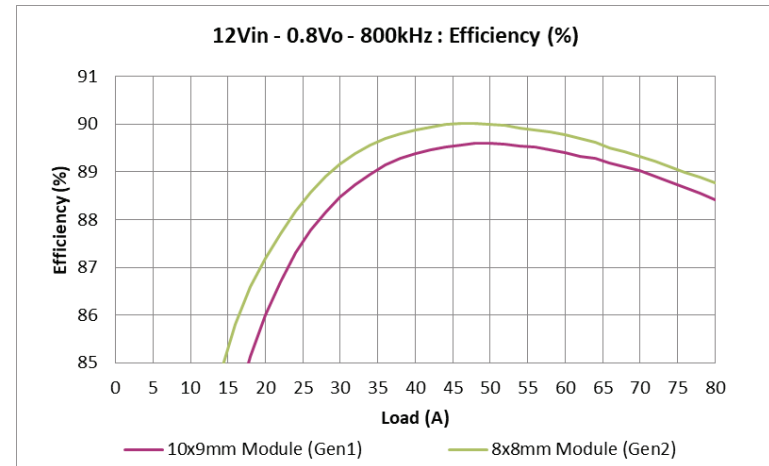


TLVR Module*

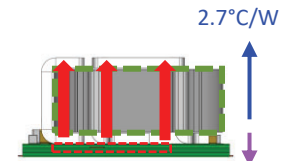
Power Density: 1.5A/mm²

- 35% smaller footprint than Gen1, same efficiency, lower thermal impedance
- Simple construction (<half the solder connections)
- Low noise due to parasitic control
- Chip embedding enables further gains in heat removal, power density, efficiency

*TLVR denotes Trans Inductor Voltage Regulator



Simple Construction

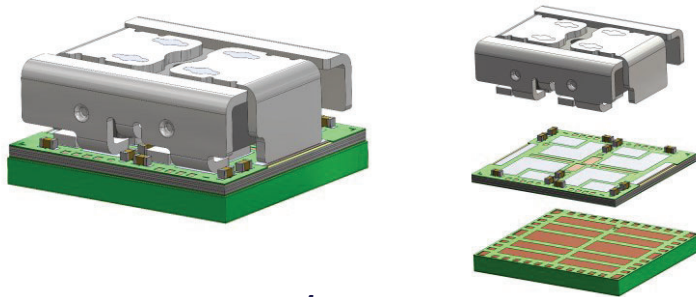


>3x reduction in junction to board thermal impedance

0.5°C/W

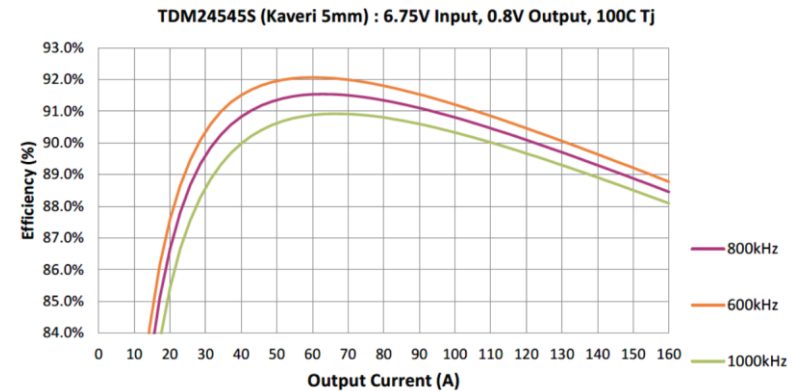
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Gen 3 Module (Vertical) : Quad Phase 160A, 10x9mm

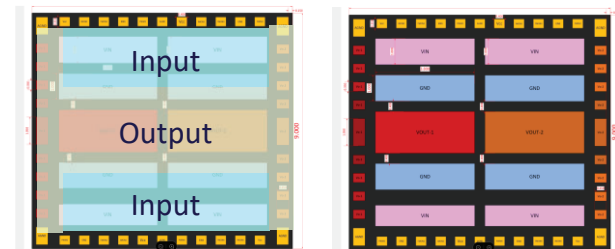


Power Density: 2.0A/mm²

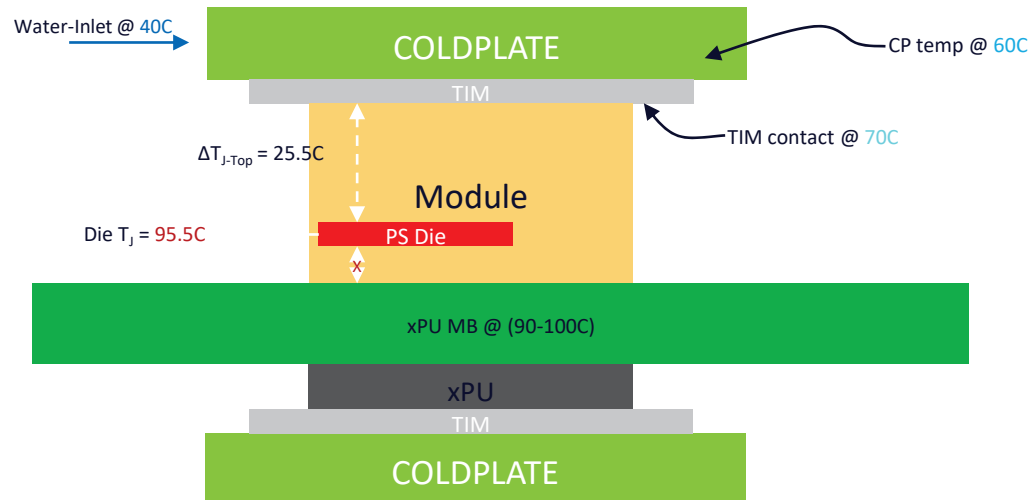
- Four phases (up to two independent outputs)
- Double power density of 1st Generation module
- Simple construction
- Capacitor version includes 40x 0402 MLCC capacitors (20x input, 20x output)
- 44uF input capacitance
- 440uF output capacitance



Footprint optimized for VPD implementation



System Thermal Estimation



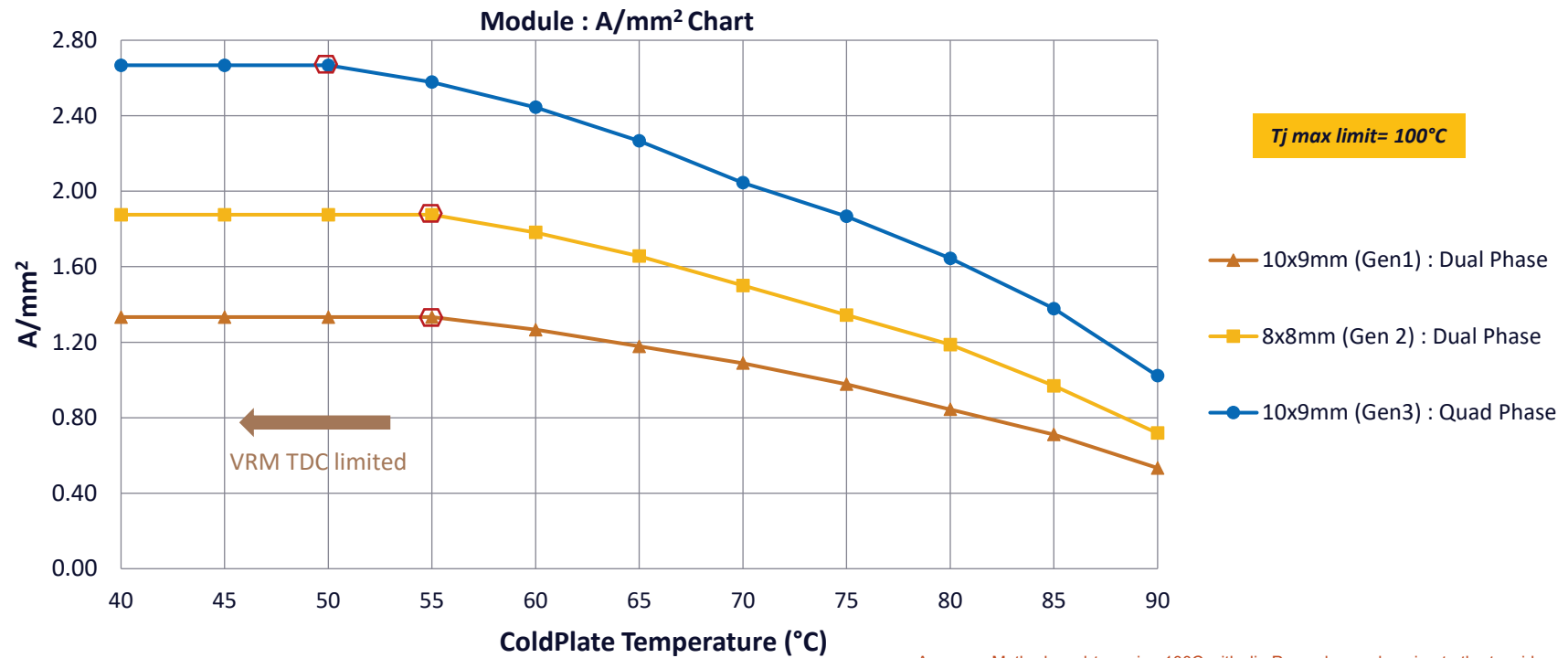
Choice of TIM: Fujipoly Sarcon GR130A (13W/mK), 0.5mm thickness

Module Thermal Estimation in a system setup:

- > Assuming all the module P_{diss} rises to the top with GPU MB at 90-100C.
- > Thermal breakdown,
 - Cold Water Inlet temperature: 40C
 - ColdPlate temperature: 60C (worse-case)
 - TIM contact: 70C (assuming 10C temp gradient)
 - Module ΔT rise: $\Delta 25.5C$ (for 80A, two phase)
 - Die temperature: ~95.5C

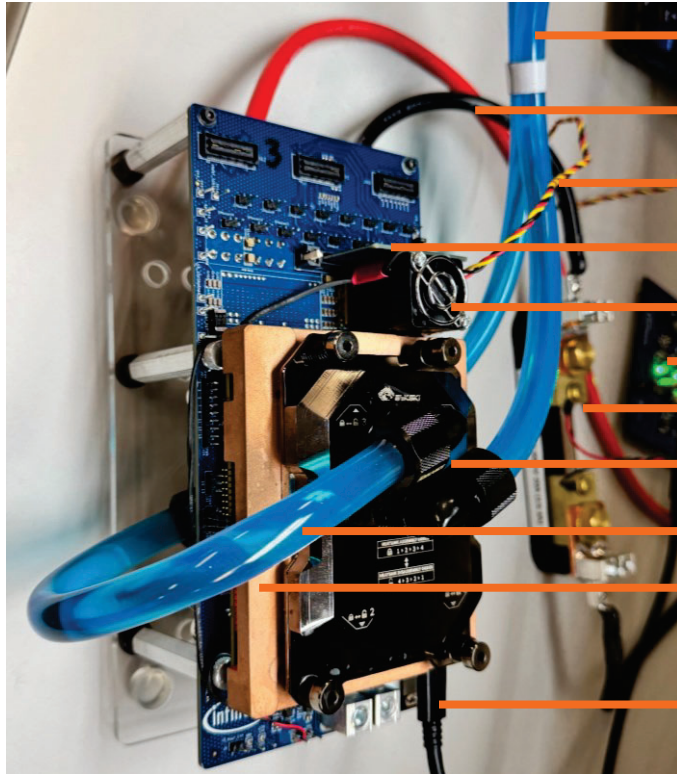
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Module – Current Density : Thermal SOA Chart (A/mm²)



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1kW Vertical Power – Test Hardware



- Inflow and outflow pipes from heat exchanger
- 54V/48V Input
- PMBus interface
- Mutiphase controller card
- 12V fan for IBC
- Load slammer control card
- Shunt to measure input current
- Load slammer cold plate connected to heat exchanger
- Top cold plate to bottom cold plate connection
- Heatsink of load slammer (underneath)
- USB interface from load slammer to control card

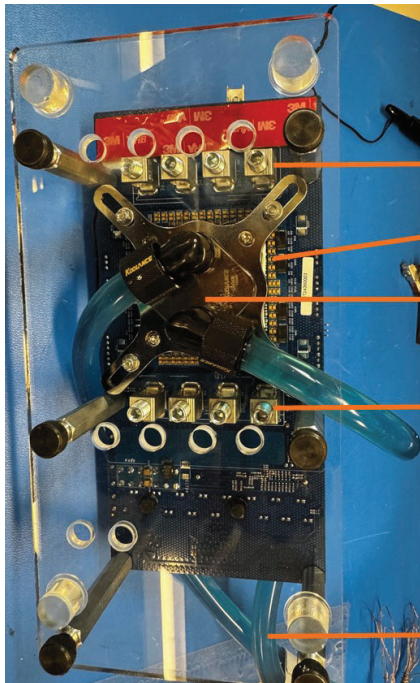


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Vertical Power Delivery – Board Setup



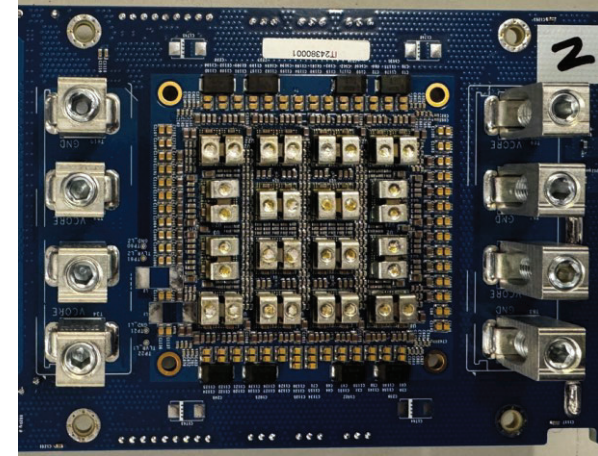
Output terminals

Modules attached underneath to cold plate

Module topside Cold plate

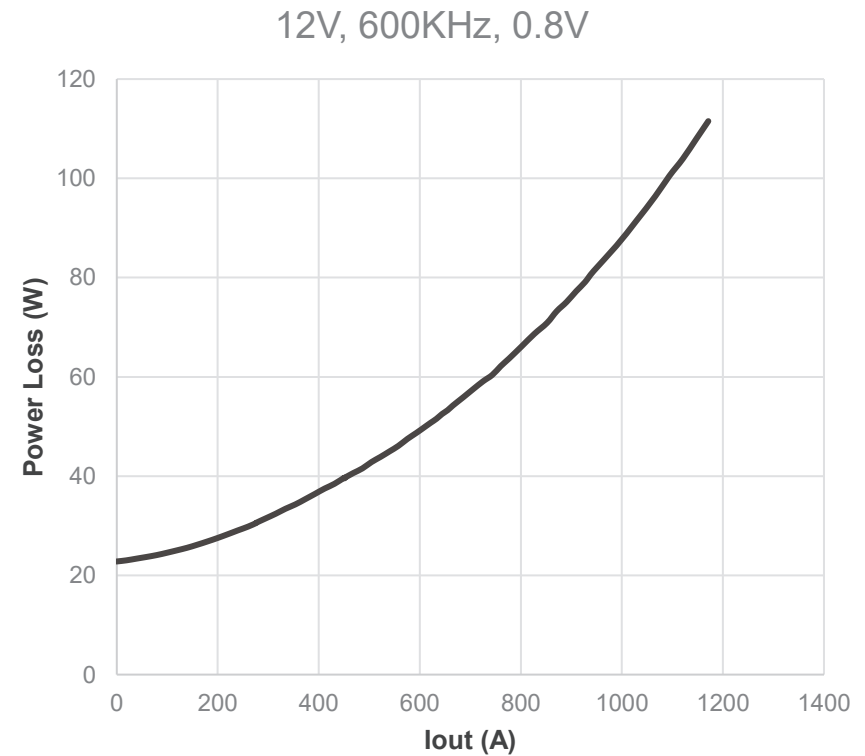
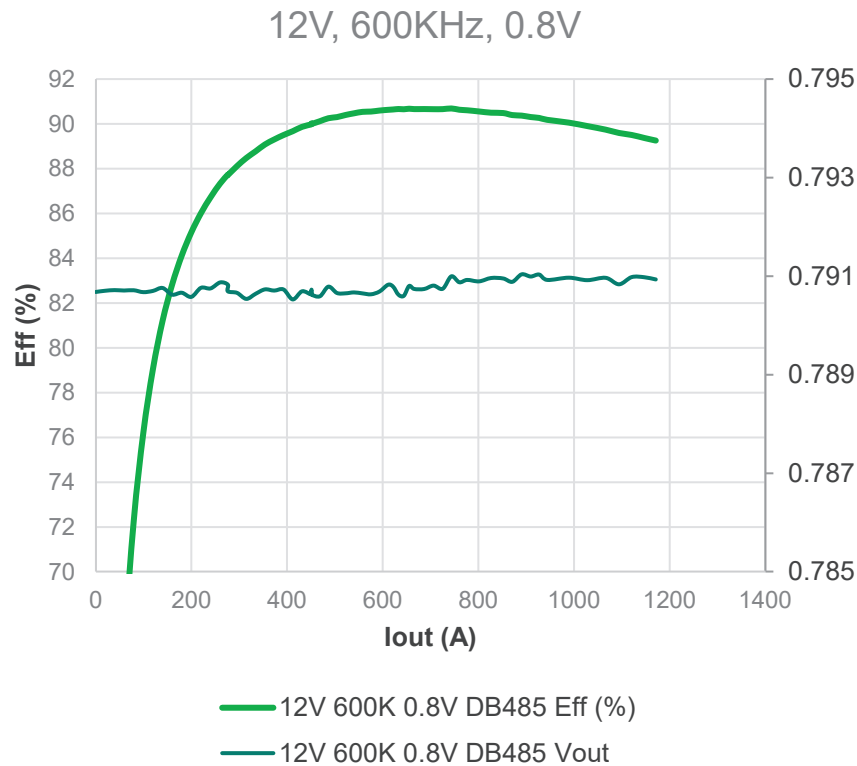
Output terminals

Inflow and outflow pipes from heat exchanger



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1kW Efficiency VPD Char Data – 12Vin to Core

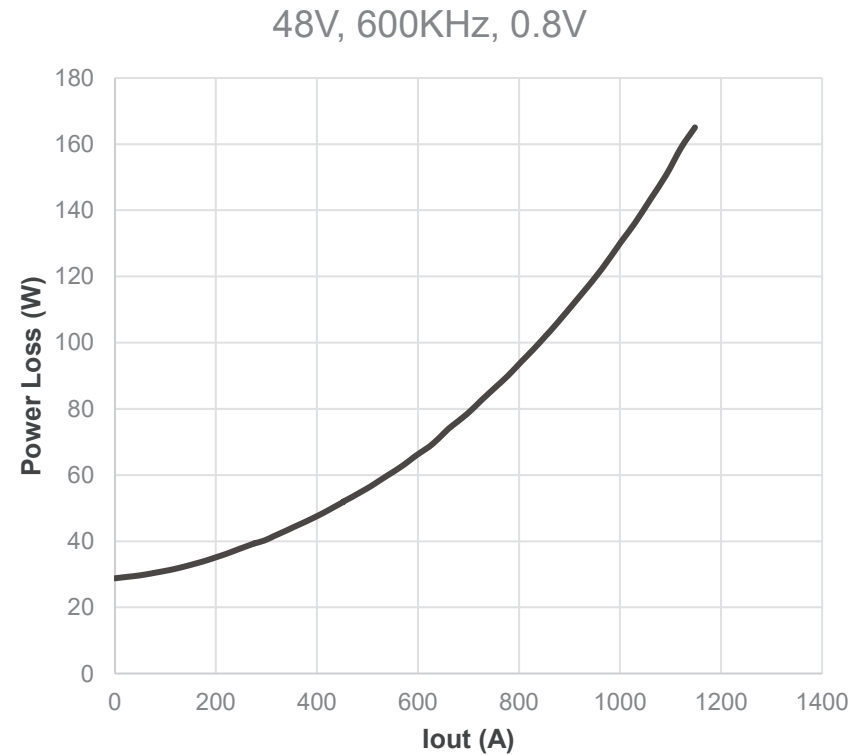
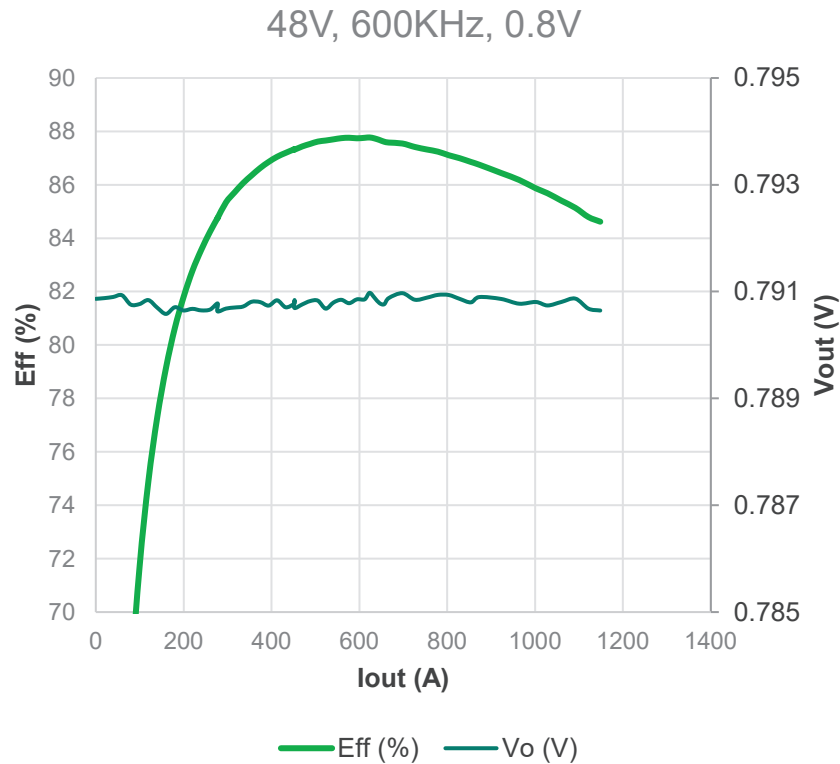


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1kW Efficiency VPD Char Data – 48Vin to Core



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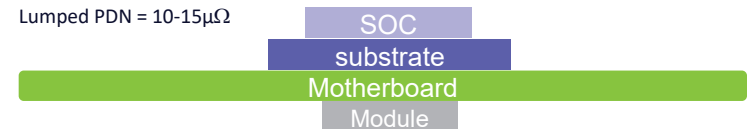
Future Progression of Power Delivery Solutions

Backside Vertical Module (BVM)

Challenges

- Increasing VRM current density to match processor requirements
- Managing motherboard and substrate PDN losses as current increases

Lumped PDN = 10-15 $\mu\Omega$



Motherboard Embedded BVM (ME – BVM)

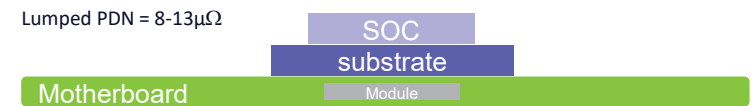
Benefits

- Bypasses motherboard, reducing total PDN by 15-20%
- Removes motherboard interconnect current limitations

Challenges

- Decreasing module profile to embed in motherboard
- Feeding power into the module and pulling heat out

Lumped PDN = 8-13 $\mu\Omega$



Substrate Embedded BVM (SE-BVM)

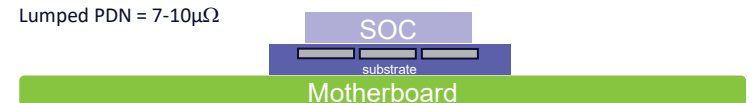
Benefits

- Reduces substrate PDN losses by additional 10-15%
- Removes substrate interconnect current limitations

Challenges

- Removing heat from module
- Limiting VR Thermal impact on processor compute capability

Lumped PDN = 7-10 $\mu\Omega$



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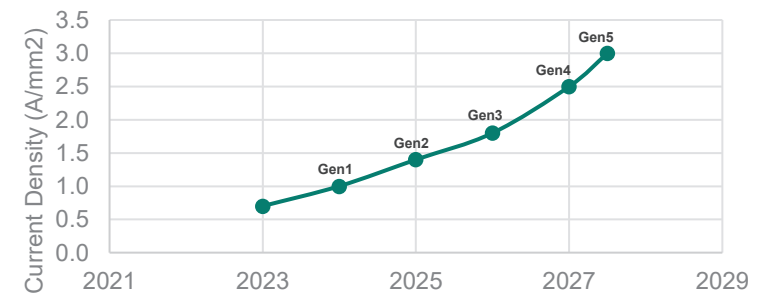
High Density Module Solutions - Summary

MP Year	<2024	2024	2025	2026
Product	Discrete	Gen 1	Gen 2	Gen 3
Footprint	4x6mm QFN	10x9x5mm 10x9x8mm	8x8x4mm	10x9x5mm
Package	CE Discrete PS	CE Discrete PS on Substrate	CE Substrate	CE Substrate w/ embedded cap
Power Density	0.7A/mm ²	1.0A/mm ²	1.5A/mm ²	2.0A/mm ²

VRM Development Goals

- Design high density scalable architecture
- Meet customer requirements for power density
- Develop technologies to address critical VPD challenges

Power Density Growth



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Call to Action

- Problem to solve
 - To standardize vertical power delivery, specifically form factor – BGA pitch – max input/output power
 - To enable an ecosystem for both module/semiconductor vendors to create complex high density vertical modules
- Ways to Contribute
 - Propose a standard form factor for high density vertical power modules
 - Develop a standard rack design to accommodate dual sided cooling
 - Identify a forum within Open Compute to continue this discussion led by a steering committee

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Thank You!



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Power module soldering and handling

Intended for TDM22544/45x

About this document

Scope and purpose

Power modules from Infineon are designed to be assembled onto PCBs using a reflow soldering process.

The purpose of this document is to provide guidelines for:

- Handling of power modules in a factory setting
- Successful design of a solder paste application and module placement process
- A module reflow profile compliant with JEDEC J-STD-020F standards
- Correct post-processing of attached power modules including rework and replacement

The recommendations in this document apply to the LG-MLGA-72-5 and LG-MLGA-72-6 packages.

Intended audience

The intended audience for this document is design engineers, technicians, developers of electronic systems, manufacturing/process engineers, and customer quality engineers of end systems using dual-phase power modules from Infineon.

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1 Packaging description

The 9 mm x 10 mm dual-phase power stage modules from Infineon consist of passive components (capacitors, inductors, resistors) and chip embedded (CE) power stages mounted on a substrate (Figure 1). The module is manufactured using lead-free solder and offered in an open frame format. The module does not contain glues or other adhesives to hold components fixed to the substrate.

The mounting surface of the module is an 8 mm x 9 mm land grid array (LGA) package with 0.55 mm square pads mounted on a 1.1 mm pitch (Figure 2). The pad finish can be electroless nickel immersion gold (ENIG) or electroless nickel electroless palladium immersion gold (ENEPIG).

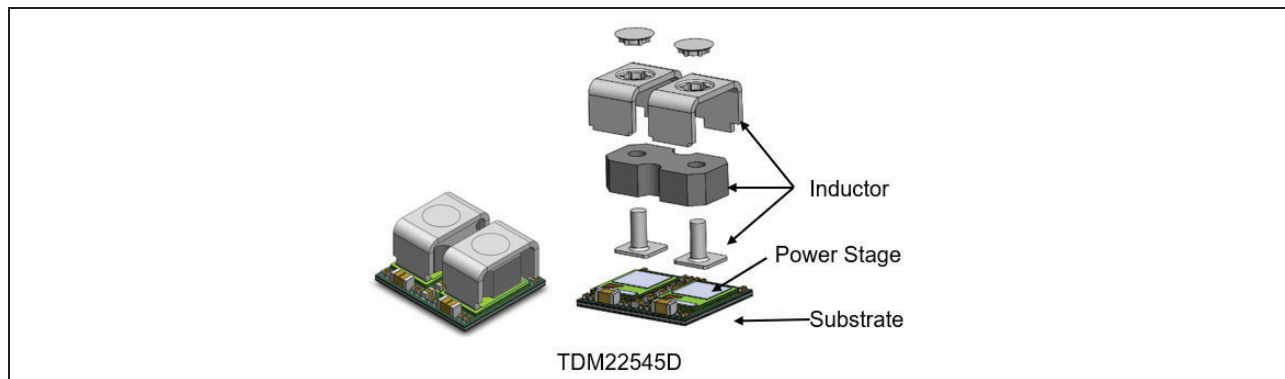


Figure 1 Dual-phase power module example (TDM22545D shown)

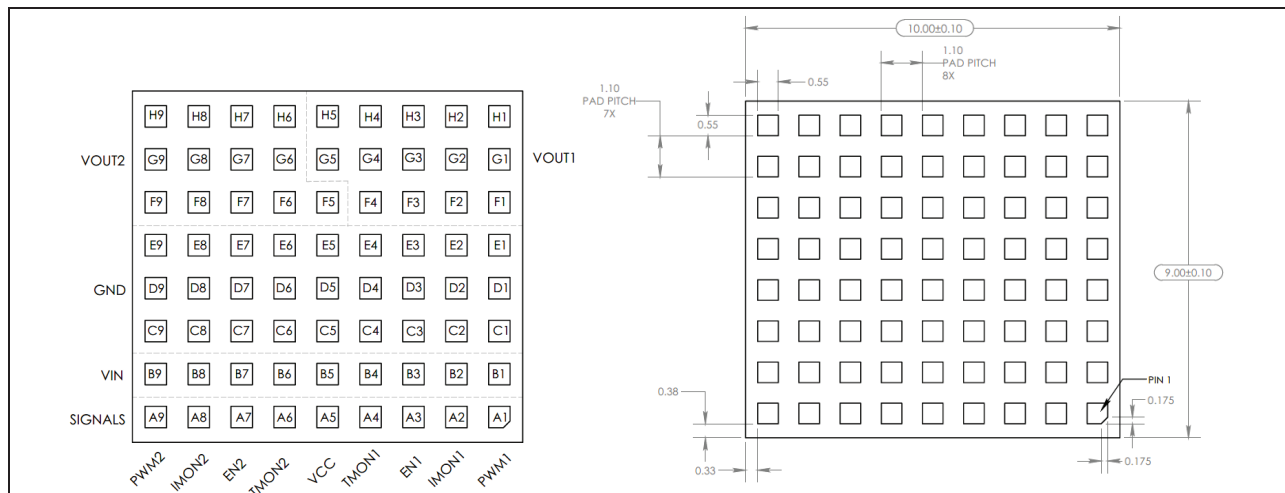


Figure 2 Transparent top view of dual phase power module LGA

2 Handling

2.1 Moisture sensitive level (MSL)

Moisture sensitivity level (MSL) ratings provide an indication of the floor life of the module after it is removed from its packaging. MSL ratings are applicable when exposed to relative humidity between 30% and 60% and temperatures between 20°C and 35°C. Unless specified on the packaging or product datasheet, power modules from Infineon are MSL 3, meaning they can be exposed to ambient conditions for up to one week without requiring baking.

Ambient exposure is cumulative if the module undergoes multiple reflow cycles, the time in between reflows all counts as time out of the bag.

2.2 General handling guidelines

- Given that the mounting LGA surface is ENIG or ENEPIG plated, the module must be kept in a vacuum-sealed bag prior to manufacturing assembly to minimize the potential for contamination.
- Do not handle modules with bare hands, as the plating will tarnish within hours after touch. Once oxidized solder will not flow on the pads.
- If the LGA pads have become contaminated after the modules have been removed from packaging, they must be cleaned before soldering. Isopropyl alcohol, a weak acid, or flux without solder may be used to clean the pads.

3 Solder process

3.1 Stencil design

Proper solder stencil design is important to ensure that the appropriate volume of solder is deposited onto the LGA pads before module placement and reflow. Infineon recommends the parameters shown in the following table for stencil design.

Table 1 Solder stencil design

Parameter	Recommendation
Stencil thickness	0.13 mm
Aperture ratio	1:1 (0.6 mm x 0.6 mm)

In addition to the appropriate stencil design, ensure to keep the stencil clean to eliminate residual solder being transferred to another board once the stencil is lifted. Applying a nano-coating to the stencil will improve release once solder has been placed. Following the print of solder, the solder should appear uniform and should fully cover the LGA pad. Use SAC305 no clean, SMQ75 indium, or other equivalent solder paste. If desired, a 0.13mm thick stencil with no-clean solder paste containing flux can be used to provide some etching of the pads during the pre-heat cycle. If flux is used, the module must be reflowed onto a board, not in an open air. Heat the flux while in a closed space to be effective.

3.2 Stencil design for LG-MLGA-72

Stencil design for LG-MLGA-72-5 and LG-MLGA-72-6 packages. A recommended stencil design is shown below.

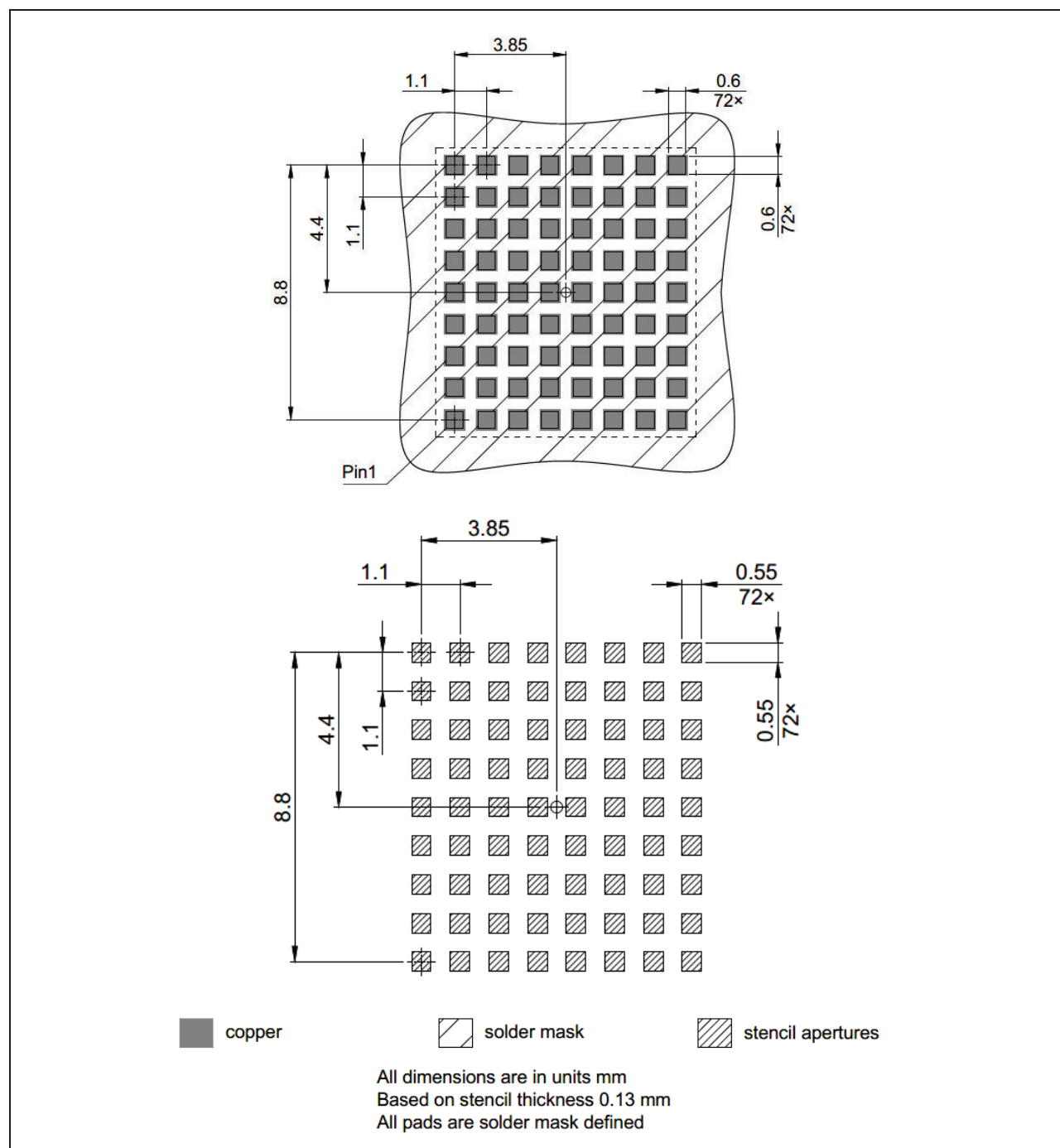


Figure 3 Stencil design

3.3 Pick and place

Design the pick and place nozzle to make a good seal with the module top surface. The dual-phase modules have distinct top inductor surfaces that can be used with a dual pick and place nozzle as shown in the figure below.

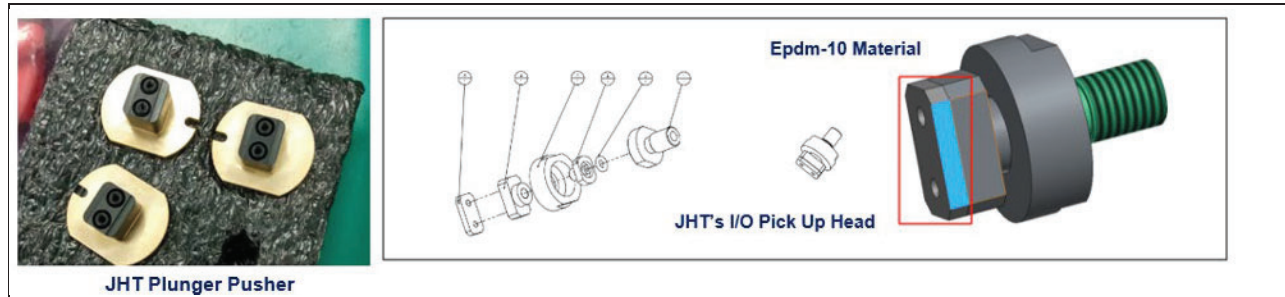


Figure 4 Pickup head design for dual-phase modules

Placement force should be high enough to ensure that all module pads get in touch with the solder paste, but low enough to not squeeze the solder paste out of the pad area. Typical values are 1 to 2 N.

The pick and place mechanism will vary based on what equipment is used to handle the modules. The following figure shows an alternate pick-up nozzle design.

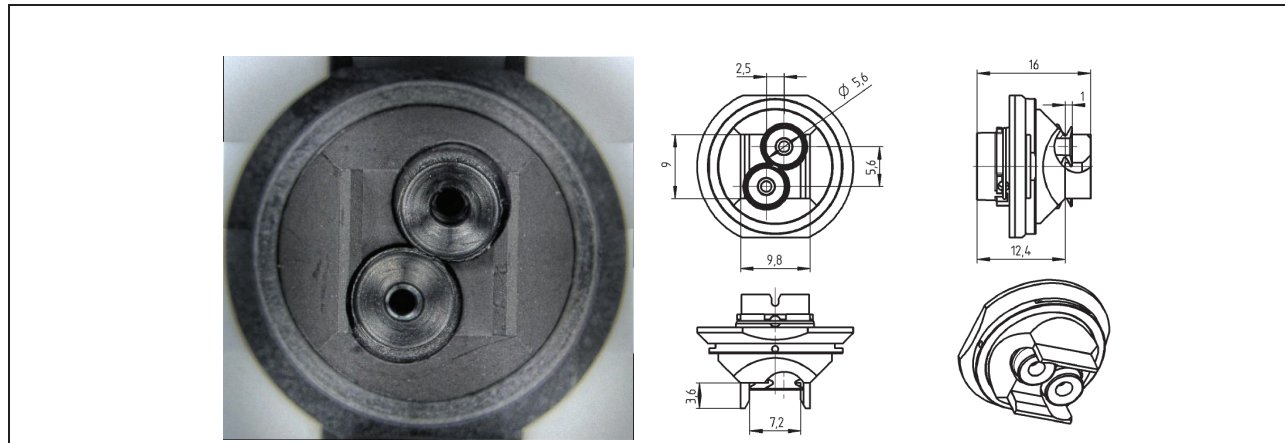


Figure 5 Alternate pick-up and place mechanism

4 Reflow

4.1 Profile

Modules are qualified for three 245°C reflows. Each reflow operation must follow the JEDEC J-STD-020D standard to form proper solder joints and minimize stresses on the module body. Three reflow operations allow for an initial mounting reflow, a second mounting reflow for additional components on a customer board and a rework reflow. The following figure shows an example reflow profile and [Table 2](#) provides soldering parameters.

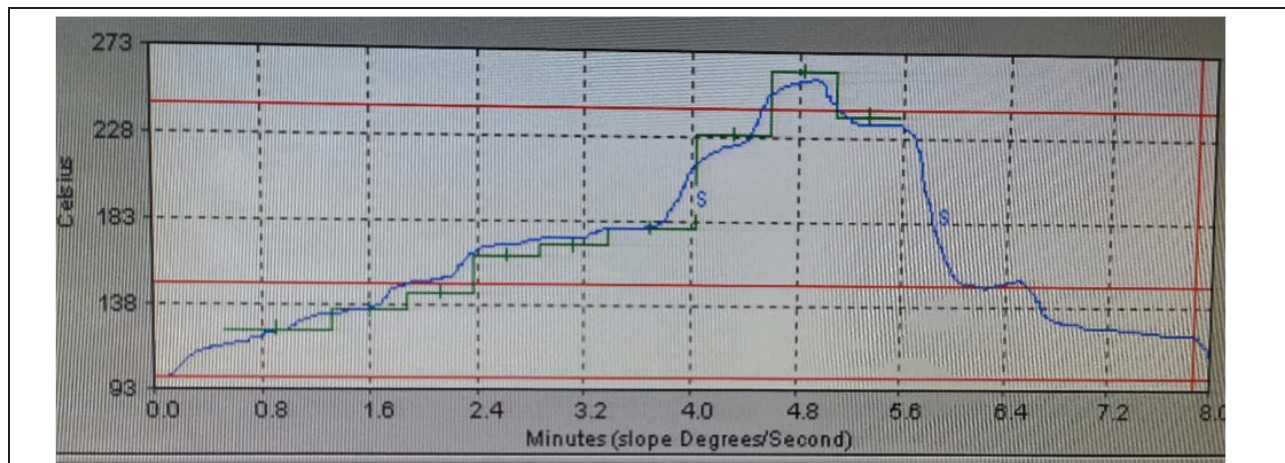


Figure 6 Reflow profile

Table 2 Reflow parameters

Parameter	Values	Notes
Classification Temperature (T_C)	245°C	Per Section 7.1, Table 2 JEDEC J-STD-020F. Module qualified with 260°C.
Peak allowable temperature (T_P)	255–260°C	Follow solder paste manufacturer guidelines for T_P .
Time above liquidus (TAL)	60-150 seconds	Maximum conditions for the component as defined per JEDEC J-STD-020F. Follow solder paste manufacturer guidelines for TAL.
Liquidus temperature	>217°C	Solder begins to soften at lower temperatures.
Preheat time	95–115 seconds	Preheat time enables flux to activate and etch LGA pads to improve adhesion and solder joint integrity.
Preheat temperature	100–150°C	–
Maximum ramp up rate (T_L to T_P)	3°C/sec	Maximum conditions for the component as defined per JEDEC J-STD-020F. Follow solder paste manufacturers recommendation.
Maximum cooling rate	6°C/sec	Maximum conditions for the component as defined per JEDEC J-STD-020F. Follow solder paste manufacturers recommendation.

The solder paste manufacturer will publish detailed parameters to optimize the reflow process for that particular paste chemistry. These parameters should be used when generating the initial reflow profile for module soldering. The profile should be further optimized for the specific board size, number of layers, copper thickness and mass of adjacent components. Please consult with Infineon if the desired reflow profile exceeds the parameters in Table 2 for T_P , TAL or maximum heating/cooling rates to ensure that the module solder integrity will not be harmed during the attach process.

4.2 Additional guidelines and recommendations

The following guidelines apply at any time the modules are undergoing reflow temperatures during assembly:

- Do not apply pressure to the top of the module during reflow. Module solder joints liquefy during the reflow process and pressure may cause solder or components to move in such a way as to wound or damage the module.
- Keep acceleration or vibration applied to the module to less than 0.5 g once the temperature exceeds 200°C. Beyond this temperature, the module solder joints will be liquidus, and excessive acceleration or vibration could cause parts to fall off.
- To the extent possible, design the reflow process to heat and cool the module uniformly. Depending on other components that can be located in close proximity to the module, this requires a faster or slower belt speed.

5 Post initial attach reflow

5.1 Inspection

Visual inspection of LGA solder joints located in the center of the module is not possible, therefore X-ray inspection is recommended for the pad connections underneath the module. The solder connections must comply with the latest version of IPC-A-610. In addition, given that all module solder connections have liquefied during the reflow, the module should be visually inspected to ensure that the module solder joints remain intact and components mounted to the module have not moved or shifted. If the guidelines in Section 4 are followed, the module solder joints will remain undisturbed following the reflow process. Any signs of solder bridging, bulging, or excessive solder balls (>100um in diameter) indicate that the module has been compromised by the reflow and the process parameters should be reviewed and optimized as needed.

5.2 Subsequent reflows

Infineon modules are qualified for up to three reflows. All three reflows should comply with the parameters shown in Table 2, however following the initial reflow to mount the modules, the subsequent reflows are focused on solder joint preservation rather than solder joint formation.

Modules may be upside down for subsequent reflows to enable double sided board assembly. In an assembly process that includes an upside reflow for modules the following guidelines should be followed.

- TDM22545x modules (5mm height) may be reflowed without support, i.e. no fixturing or glue is required to hold the module in place while being reflowed upside down. In this case, while the module solder joints are liquidus during the process, solder surface tension counteracts the weight of the module and components, holding them in place.
- If support is used during the upside down reflow process, care must be taken to ensure that pressure is not applied to the module, as this may cause solder or components to move. Any force that is applied should not exceed 1.24gf (the approximate weight of the inductor clip) to insure no significant movement occurs during the time joints are liquefied.
- As mentioned in section 4.2, keep acceleration or vibration applied to the module to less than 0.5 g once the temperature exceeds 200°C to minimize component movement.
- While the focus of the reflow profile will be the components being mounted, the ramp and cool down rates of the module during a subsequent reflow should be monitored closely to ensure that module solder connections are not compromised by the process.
- Visual inspection of module solder joints per section 5.1 is recommended following subsequent reflows.
- The module solder joints will not become liquidus at temperatures below 200°C. Any reflow profile that does not expose the module to temperatures above 200°C does not count as a reflow for the module and does not need to follow these guidelines.

5.3 Rework

If needed, a module can be removed from the main board by heating it above 220°C with a hot air gun and then lifting the module off with tweezers. Once removed in this way, the module must be discarded as internal components and solder have likely shifted during this process. During the application of heat above 220°C, ensure that care is taken as to not induce any pressure or movement in adjacent modules during the rework process.

A module may be replaced by cleaning the main board pads, re-printing solder, and repeating the process outlined in this document.

6 Mechanical Considerations and Heatsink Mounting

The module is designed to efficiently conduct the heat away from the silicon dies inside the power stages both from the top side of the module via the copper clips and through the bottom side of the module via the PCB.

As shown in the picture below, the thermal energy can be efficiently extracted from the module by attaching a heat sink or a cold plate on the top side of the module. A thermal interface material (thermal gap pad or thermal putty) between the module and the heat sink/cold plate helps improve the thermal path by reducing the interfacial contact resistances. The thermal interface material datasheets and application notes suggest recommended mounting pressure for lower thermal resistance. Infineon recommends mounting pressure to be ≤ 100 psi; exceeding this mounting pressure may potentially damage the module and may also cause parametric shifts in performance.

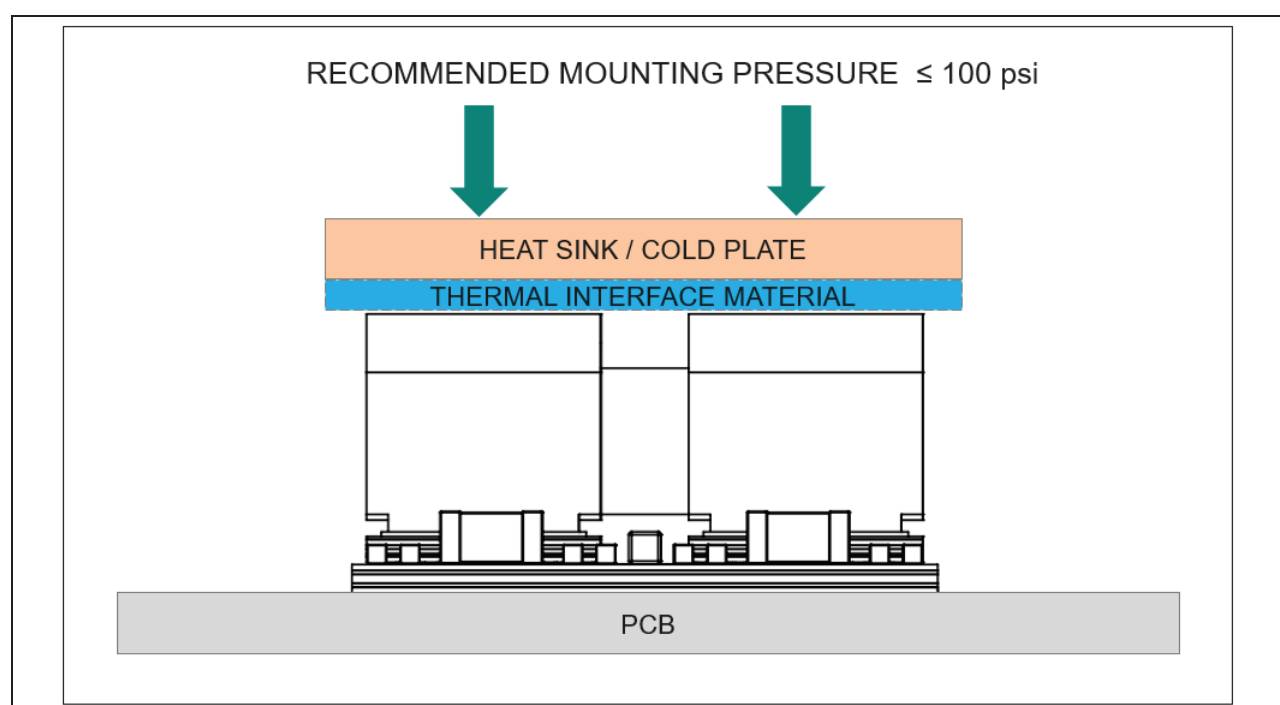


Figure 7 Recommended mounting pressure

If part of the system assembly, Infineon modules should have the heatsink or coldplate mounted to them post reflow. When mounting a heatsink, the following guidelines should be observed.

- Maximum vertical pressure applied to the module should be kept below 100psi.
- Applying force to the module off vertical axis should be avoided. In particular off axis force applied to the inductor body can apply shear stress to the solder joints located under the power stages on the modules, causing fracture, intermittent contact and module failure.
- When applying vertical force to the module, the back side of the customer board should be supported to prevent board deflection. Deflection can cause shear stress to solder joints and cause compromise and failure.

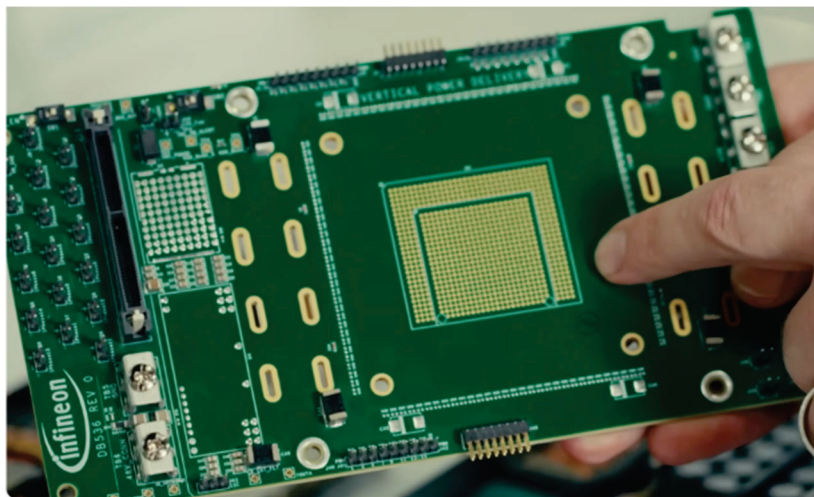
- In double-sided assemblies, the board must be supported when mounting heatsinks or other mechanical components on the opposite side to prevent board deflection damage in the opposite direction.
- Either compliant pads or dispensable Thermal Interface Materials (TIM) can be used to couple module heat to the heatsink. When using a TIM pad, the module tolerance stack should be considered to properly size the pad thickness. When using a dispensable TIM, the dispense method should ensure 100% coverage of the module top surface.
- The metal top surface of the module is at V_{OUT} potential. The mechanical stack should be designed such that no casual contact between grounded heatsinks and top clip is possible in the worst case. Given the low potential of V_{OUT} , most TIMs have sufficient resistance so as to prevent current flow from clip to heatsink. The resistivity of the TIM should be checked to confirm no significant current will flow in the system.

References

- [1] Infineon Technologies AG: *TDM22544D webpage*; [Available online](#)
- [2] Infineon Technologies AG: *TDM22545D webpage*; [Available online](#)

Revision history

Document revision	Date	Description of changes
V 1.0	2024-04-19	Initial release
V 1.1	2024-06-04	Section 3.2 added. Recommended stencil design for LG-MLGA-72 Section 6.0 added. Mechanical recommendations
V 2.0	2025-03-31	Updated to include guidelines for subsequent reverse side reflow



Infineon Talks Powering AI and Infrastructure

[Home](#)[Technology news](#)[2025](#)[Infineon and Delta collaborate on high-density power modules to accelerate data center power architecture](#)

Infineon and Delta collaborate on high-density power modules to accelerate data center power architecture

Technology news • Aug 28, 2025

Munich, Germany – 28 August 2025 – Infineon Technologies AG (FSE: IFX / OTCQX: IFNNY) today announced the strengthening of its existing collaboration with Delta Electronics, Inc., in the development of state-of-the-art high-density power modules capable of enabling vertical power delivery to AI processors in hyperscale data centers. Together, the companies are taking a next step to drive decarbonization and digitalization in AI data centers further.

The partnership leverages Infineon's ultra-thin silicon (MOSFET) chip technology and embedded packaging expertise, as well as Delta's industry-leading power module design and manufacturing capabilities. This results in highly dense modules with exceptional efficiency to enable vertical power delivery (VPD) architecture to the xPUs in hyperscale data centers. The usage of vertical power delivery modules in comparison to a lateral mounted discrete solution can save up to 150 tons of CO₂ over an expected lifetime of three years per rack. Assuming that future hyperscale data centers will consist of up to 100 server racks, the amount of carbon dioxide saved is equivalent to the CO₂ emissions of 4000 households per year.

“The close collaboration with Delta is a perfect example of complementarity – combining Infineon’s best-in-class silicon and packaging expertise with Delta’s superior module development capabilities,” said Adam White, Division President Power & Sensor Systems at Infineon. “Together, we're offering

tangible value to the hyperscalers, maintaining high efficiency and robustness while reducing cost and further driving decarbonization.”

“Our collaboration has resulted in the development of highly advanced VPD modules, which allows us to provide unparalleled power efficiency, reliability, and scalability for our customers,” said Ares Chen, Delta’s Vice President and General Manager of its Power and System Business Group. “We’re proud to be playing a key role in fostering energy conservation in the hyperscale data center power infrastructure ecosystem, and look forward to realizing significant contribution for the entire industry through our joint innovation.”

Infineon's OptiMOS™ silicon-based 90 A integrated power-stage solution is being utilized by Delta to develop VPD modules. The use of vertical power delivery is a key factor in improving system efficiency as it allows for a more direct and compact power delivery path. By delivering power vertically, rather than horizontally, the VPD modules reduce power delivery network losses in the system. This, in turn, enables the modules to achieve higher power density and efficiency, while also reducing the amount of heat generated due to less power loss. Additionally, the vertical power delivery design also frees up space on the system board, allowing hyperscalers a more efficient use of space and the development of more compact and dense data center designs reducing total-cost of ownership.

As the AI era continues to advance, the demand for high-power data centers is increasing exponentially. To meet this demand, data centers require power delivery solutions that can support the growing power needs of AI training and inferencing applications. Furthermore, the increasing densification of IT racks, driven by the integration of tightly interconnected xPUs requires the development of higher voltage power distribution solutions that can efficiently deliver power to these high-density systems. In response to these challenges, Infineon and Delta are cooperating in the development of next-generation power delivery solutions for AI data centers that will require 1 megawatt (MW) per rack in the near future.

Alongside the company’s broad portfolio in gallium nitride (GaN) and silicon carbide (SiC) power solutions, Infineon is the global leader in silicon (Si) MOSFETs that are designed to enable best-in-class performance, reliability and ease-of-use in power switching, control and conversion. This means more

efficiency, power density and cost effectiveness for products and system solutions.

Learn more about Infineon power modules for AI data centers [here](#).

About Infineon

Infineon Technologies AG is a global semiconductor leader in power systems and IoT. Infineon drives decarbonization and digitalization with its products and solutions. The Company had around 58,060 employees worldwide (end of September 2024) and generated revenue of about €15 billion in the 2024 fiscal year (ending 30 September). Infineon is listed on the Frankfurt Stock Exchange (ticker symbol: IFX) and in the USA on the OTCQX International over-the-counter market (ticker symbol: IFNNY).

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Infineon OptiMOS™ TDM22544D

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